



US008854746B2

(12) **United States Patent**
Hsu et al.

(10) **Patent No.:** **US 8,854,746 B2**
(45) **Date of Patent:** **Oct. 7, 2014**

(54) **IMAGE CAPTURING OPTICAL LENS SYSTEM**

(71) Applicant: **Largan Precision Co., Ltd.**, Taichung (TW)

(72) Inventors: **Po-Lun Hsu**, Taichung (TW);
Hsin-Hsuan Huang, Taichung (TW)

(73) Assignee: **Largan Precision Co., Ltd.**, Taichung (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 41 days.

(21) Appl. No.: **13/757,587**

(22) Filed: **Feb. 1, 2013**

(65) **Prior Publication Data**

US 2014/0177076 A1 Jun. 26, 2014

(30) **Foreign Application Priority Data**

Dec. 25, 2012 (TW) 101149767 A

(51) **Int. Cl.**
G02B 9/60 (2006.01)
G02B 13/18 (2006.01)

(52) **U.S. Cl.**
CPC **G02B 13/18** (2013.01)
USPC **359/714**; 359/764

(58) **Field of Classification Search**
CPC G02B 9/60; G02B 9/64;
G02B 13/0045
USPC 359/708–714, 754–758, 763–766
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,825,556 A	10/1998	Meyers	
8,000,030 B2	8/2011	Tang	
8,179,470 B2	5/2012	Chen et al.	
2012/0243108 A1 *	9/2012	Tsai et al.	359/713
2013/0016261 A1 *	1/2013	Tanaka et al.	348/294

FOREIGN PATENT DOCUMENTS

JP	50-26534	3/1975
JP	56-123511	9/1981
JP	58-33211	2/1983
JP	63-274904	11/1988
JP	2012-37763	2/2012

* cited by examiner

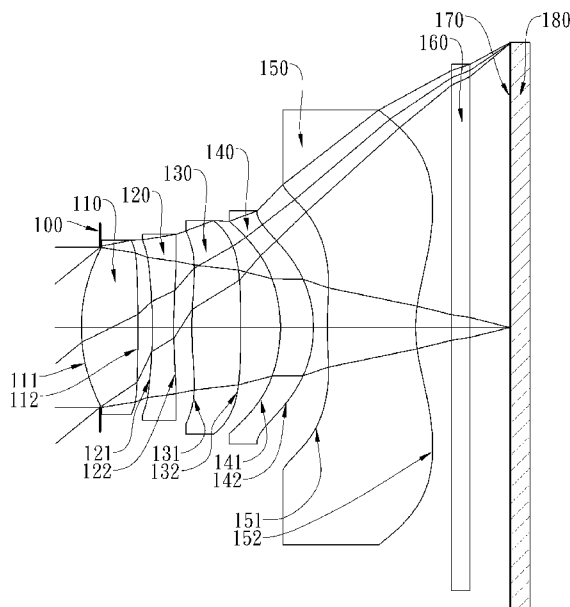
Primary Examiner — Darryl J Collins

(74) Attorney, Agent, or Firm — Tim Tingkang Xia, Esq.;
Morris Manning & Martin, LLP

(57) **ABSTRACT**

An image capturing optical lens system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The first lens element with positive refractive power has a convex object-side surface. The second lens element with negative refractive power has a concave object-side surface. The third lens element has refractive power. The fourth lens element with refractive power has a concave object-side surface and a convex image-side surface. The fifth lens element with refractive power has a concave image-side surface, wherein an object-side surface and the image-side surface of the fifth lens element are aspheric, and the fifth lens element has at least one inflection point on the image-side surface thereof.

23 Claims, 18 Drawing Sheets



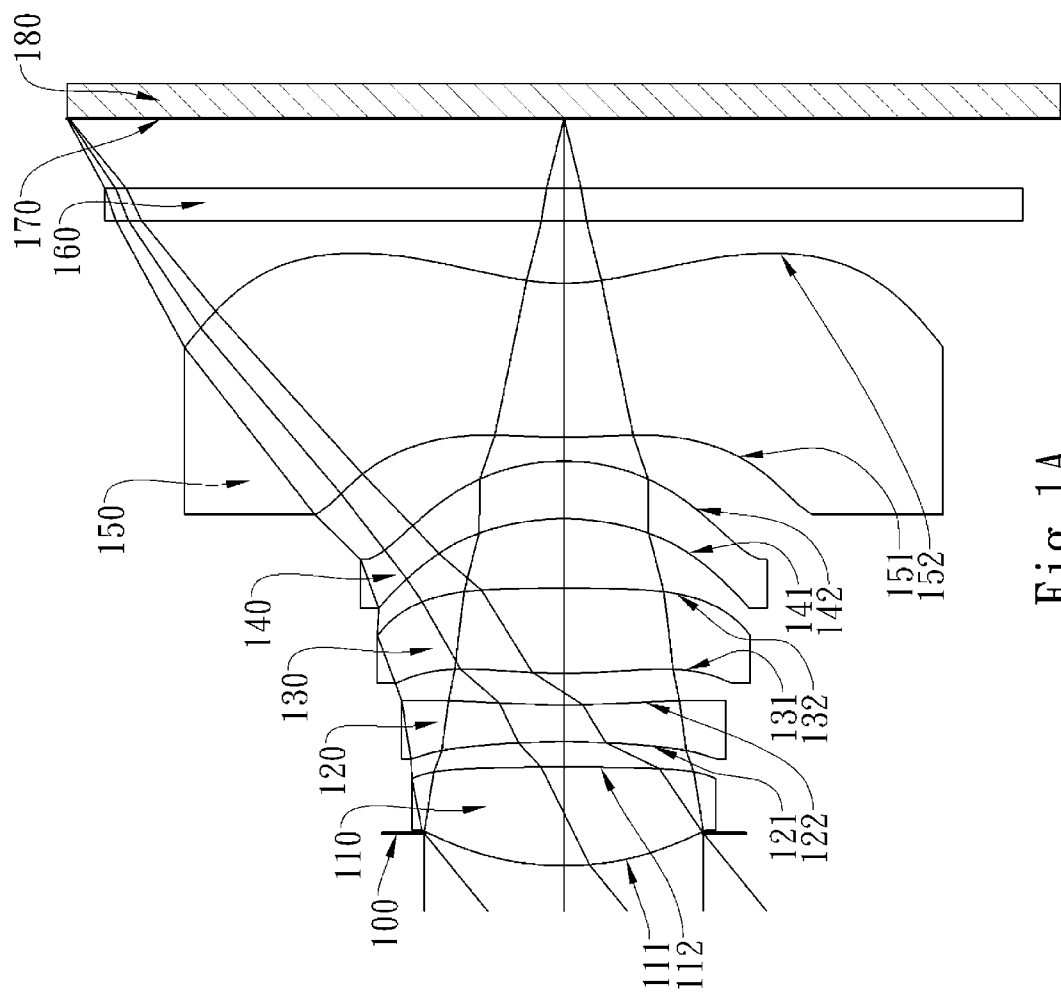


Fig. 1A

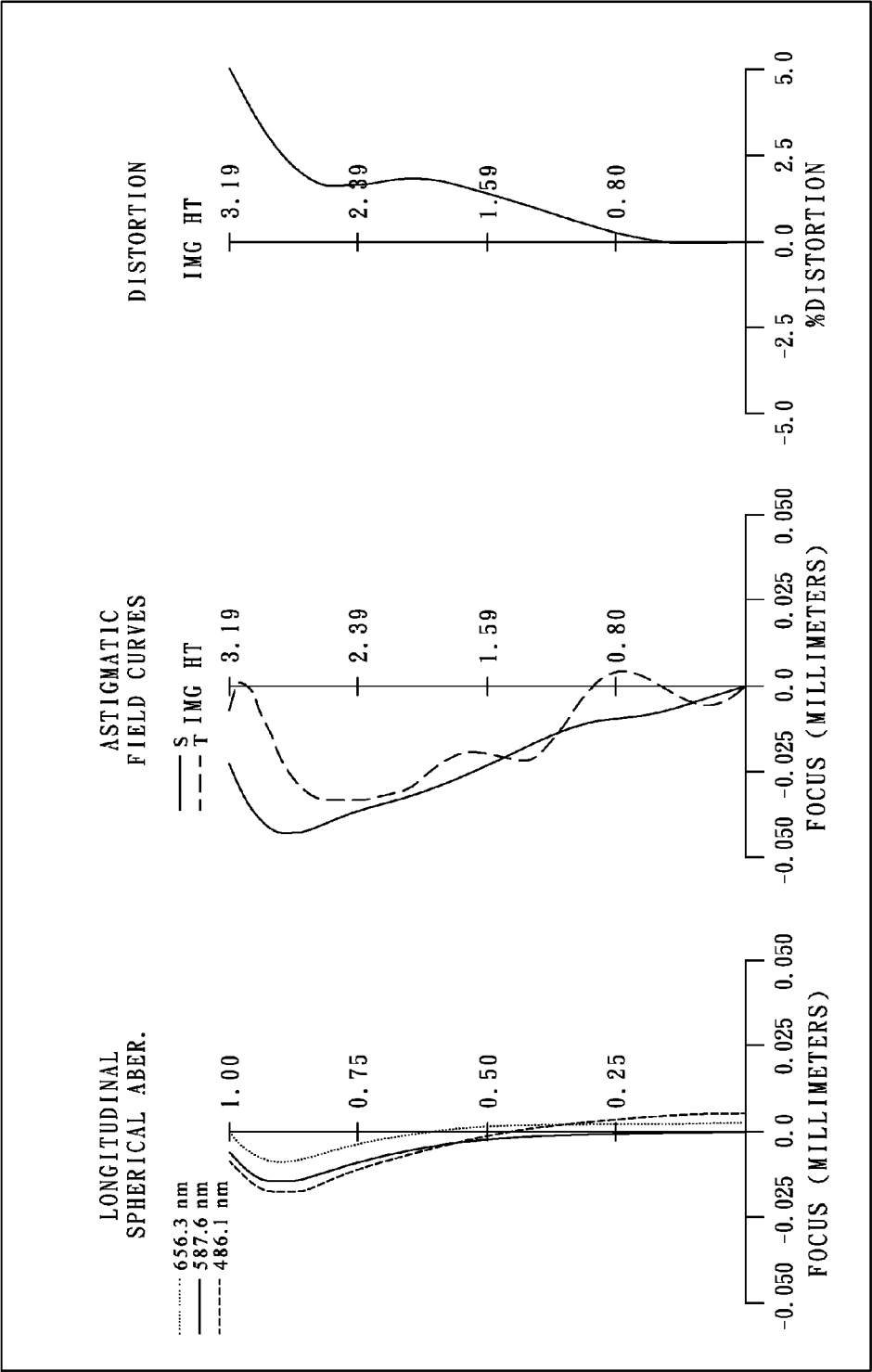


Fig. 1B

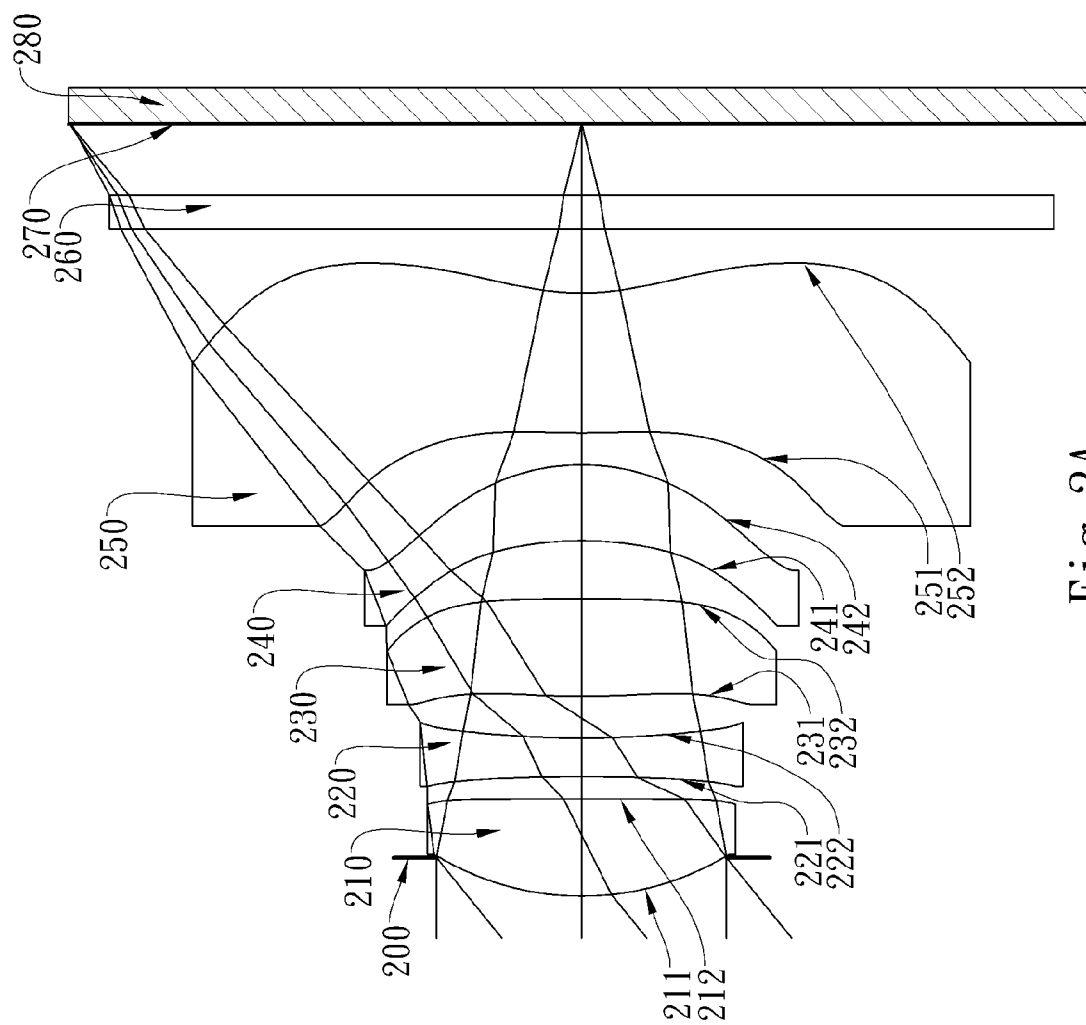


Fig. 2A

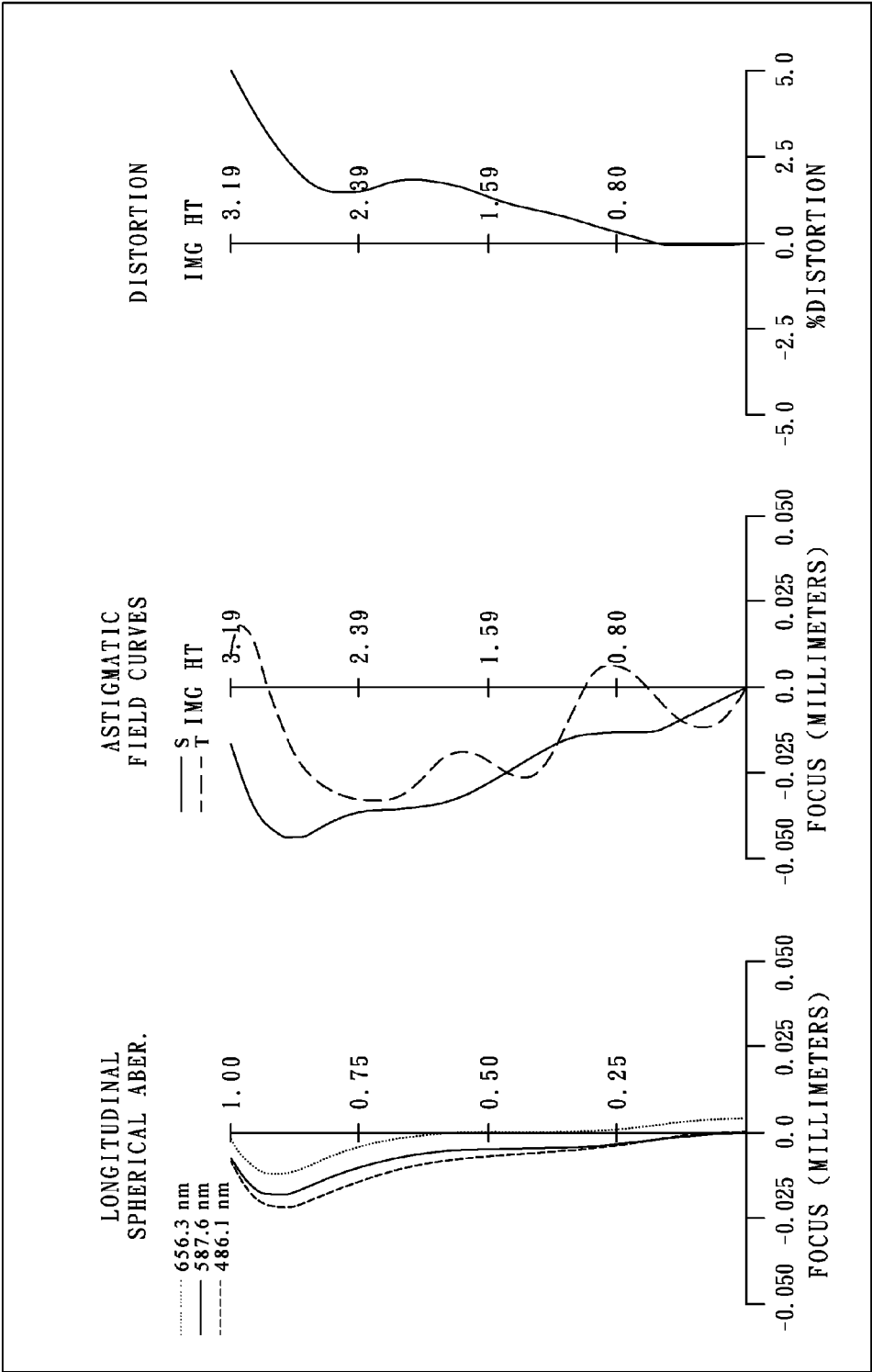


Fig. 2B

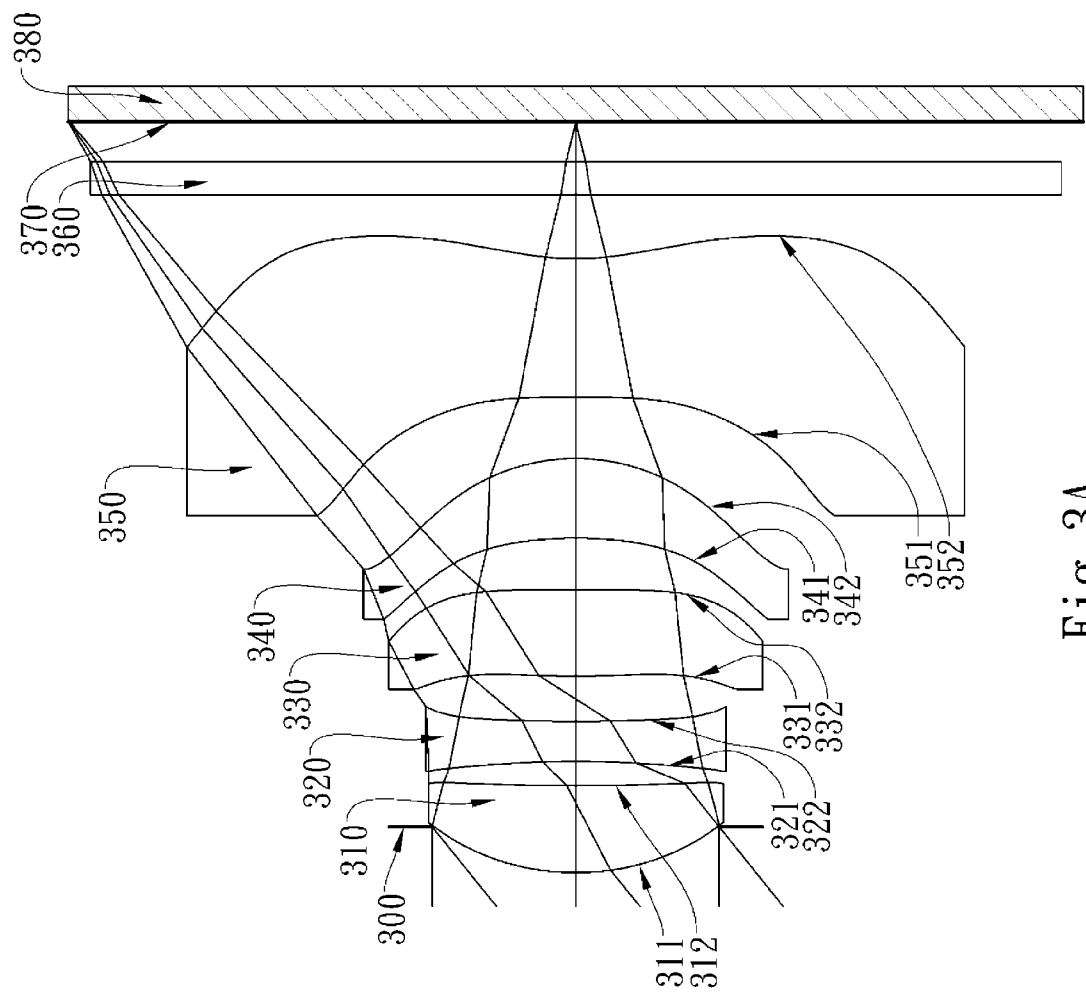


Fig. 3A

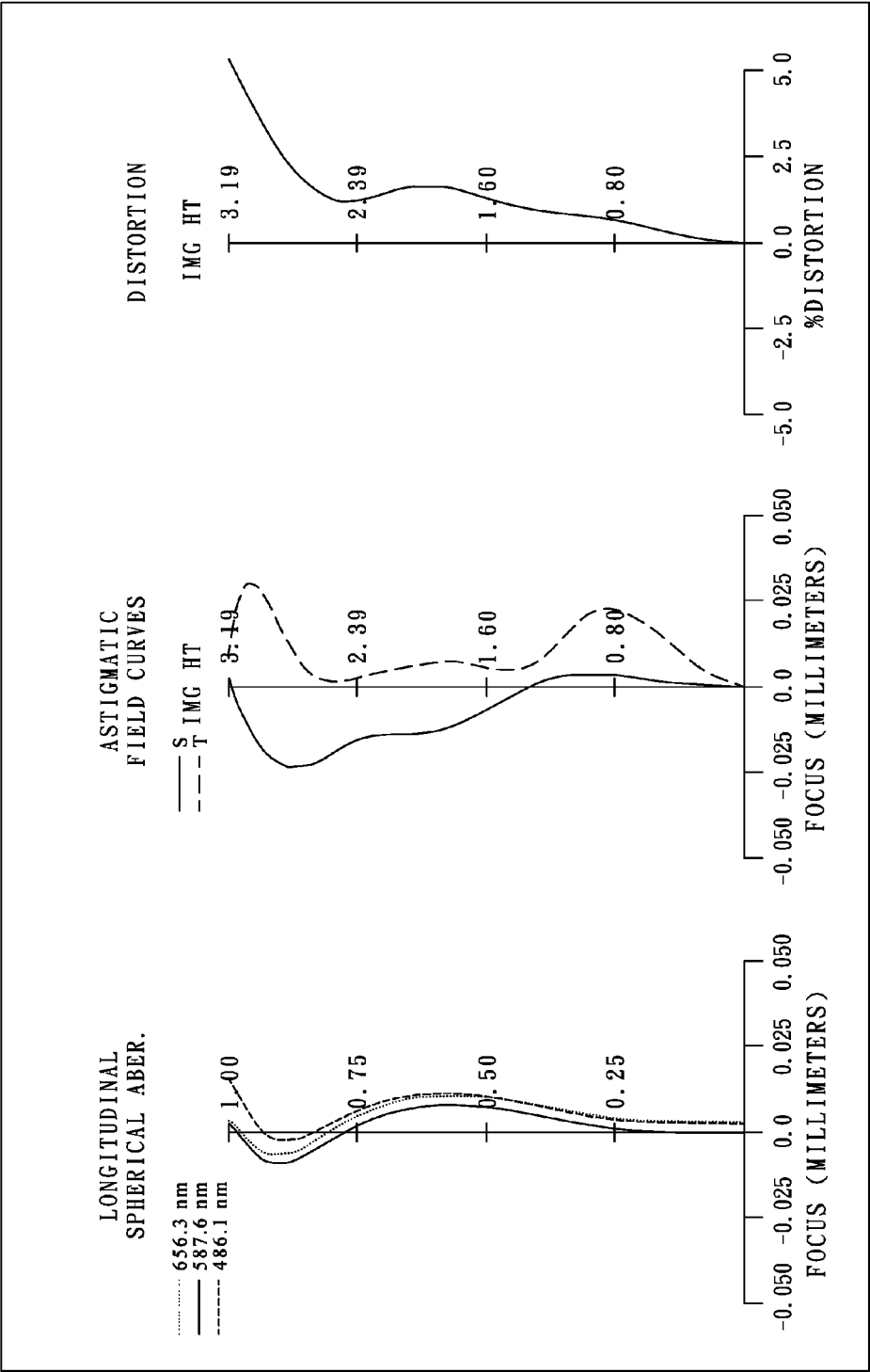


Fig. 3B

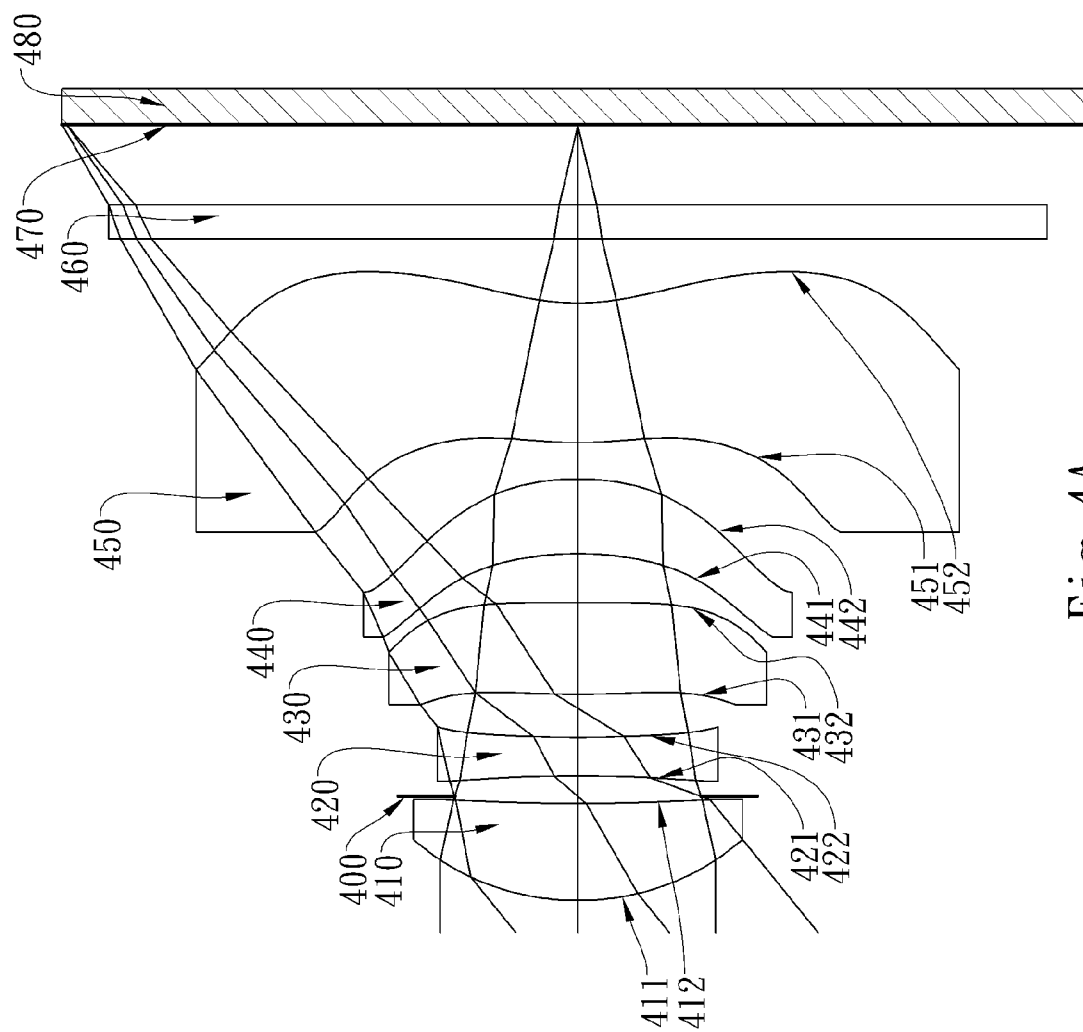


Fig. 4A

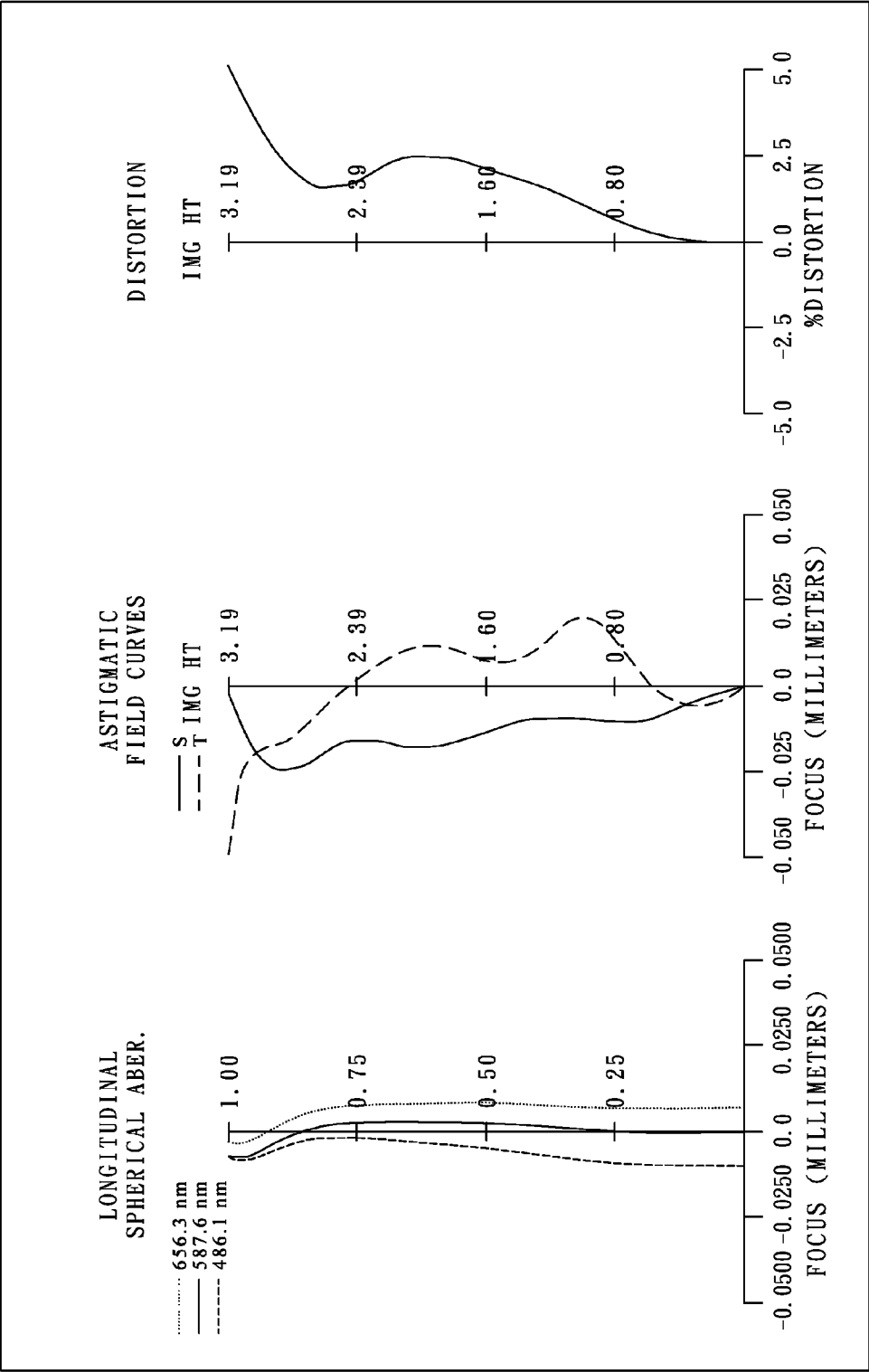


Fig. 4B

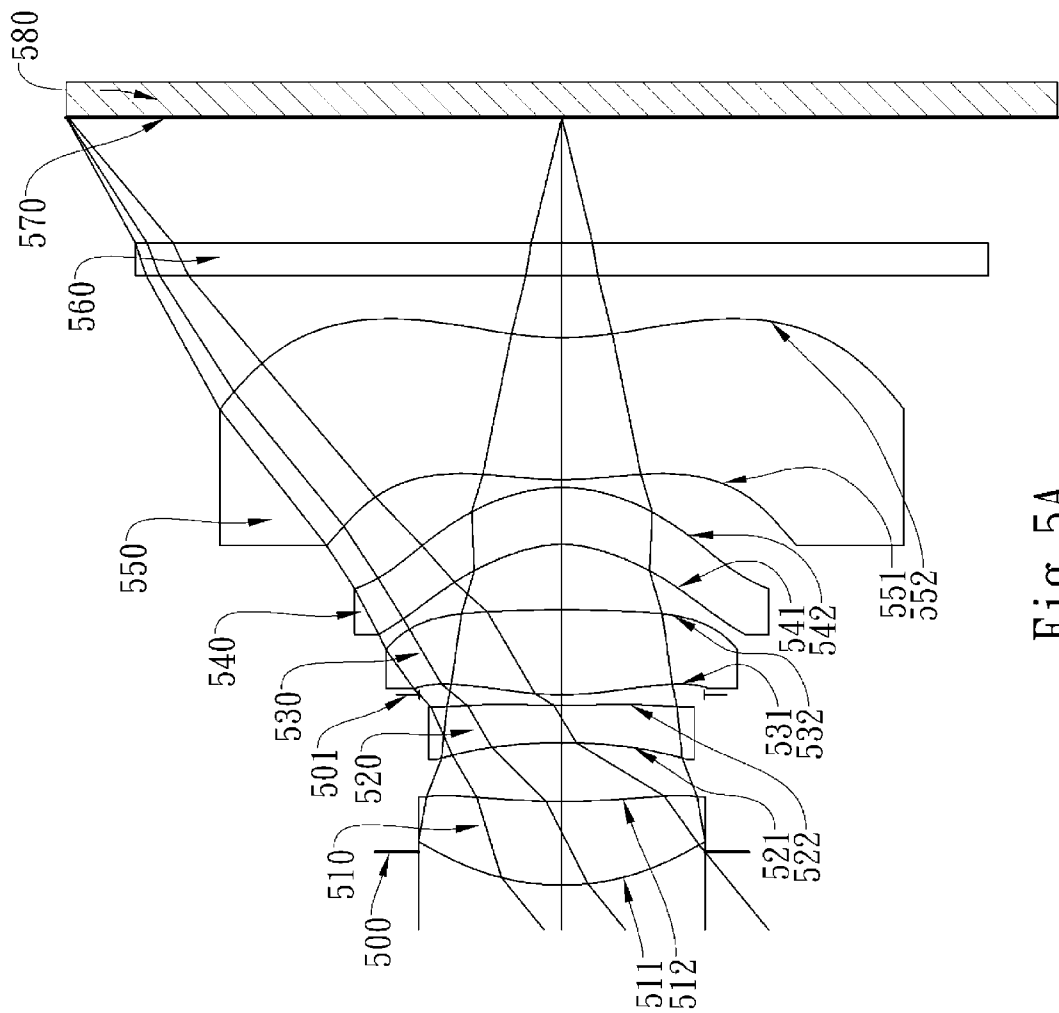


Fig. 5A

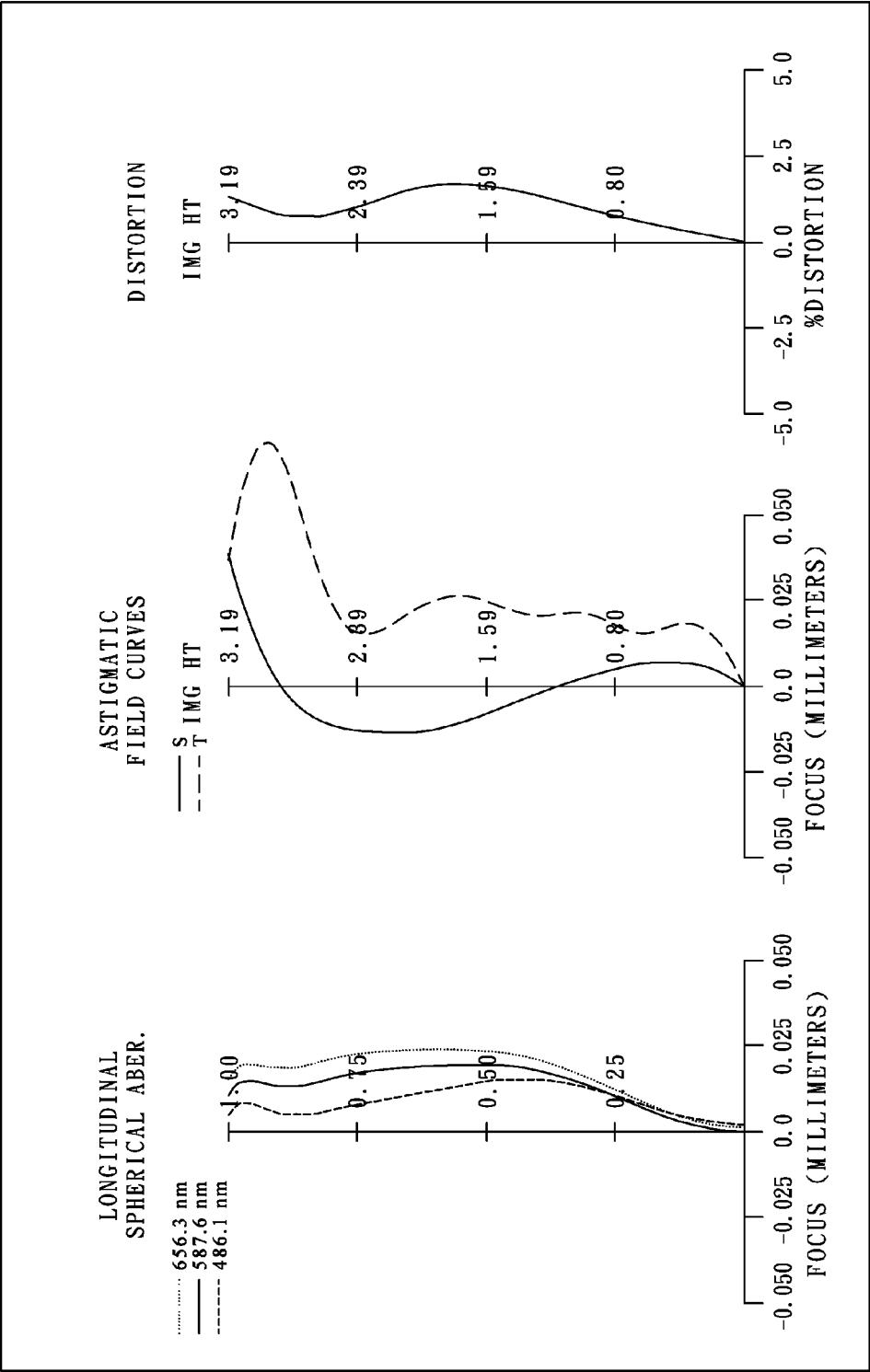


Fig. 5B

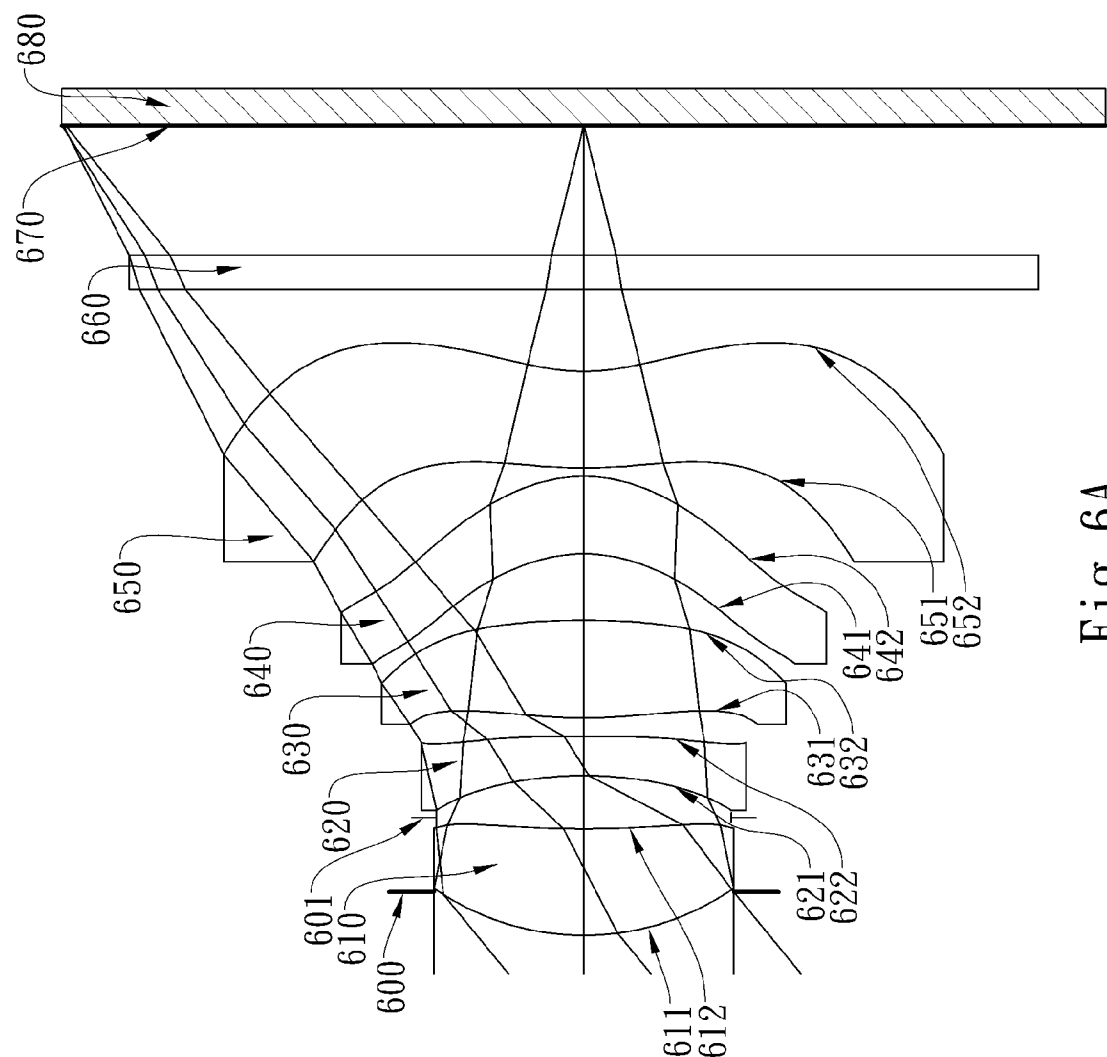


Fig. 6A

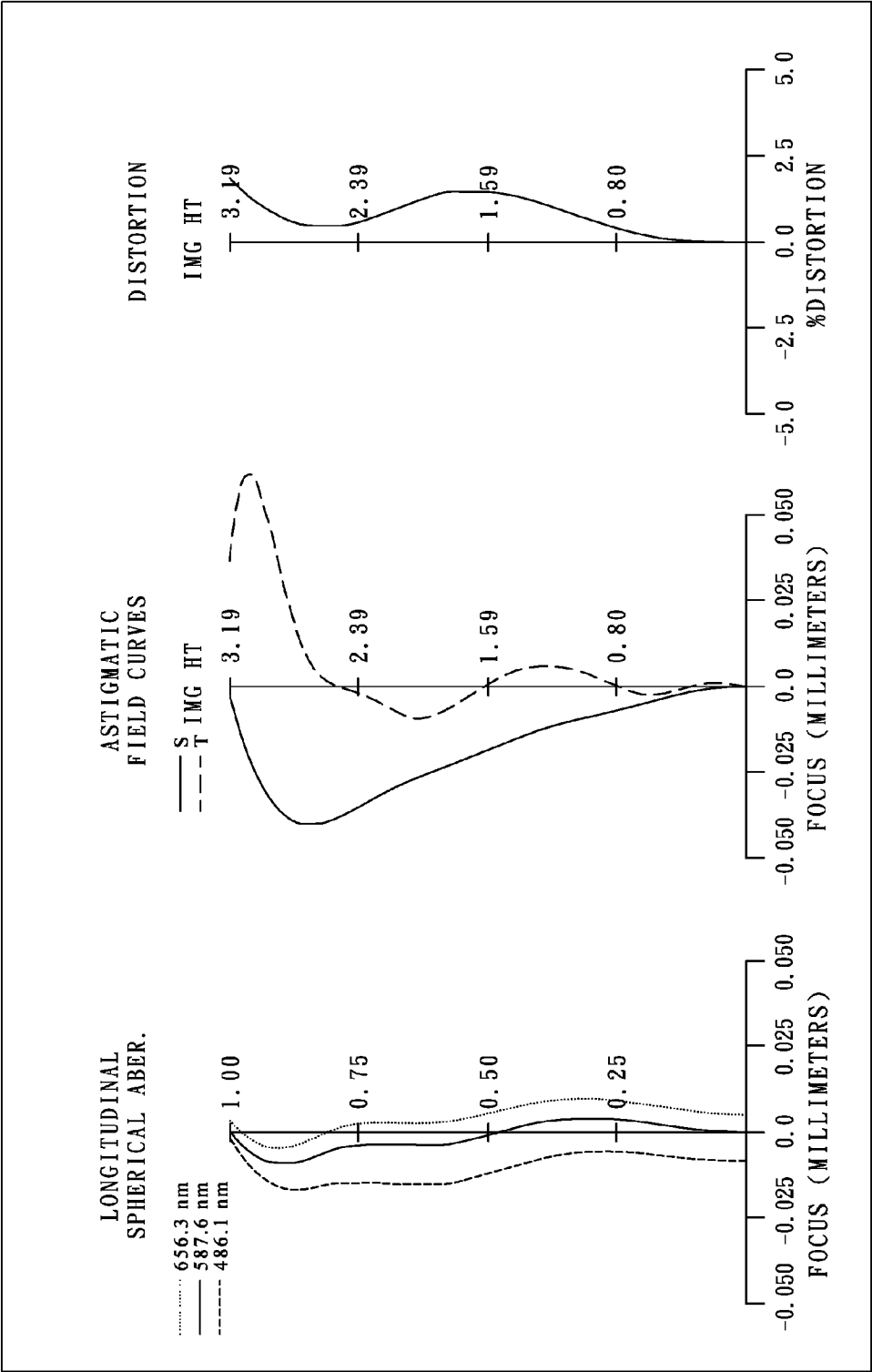


Fig. 6B

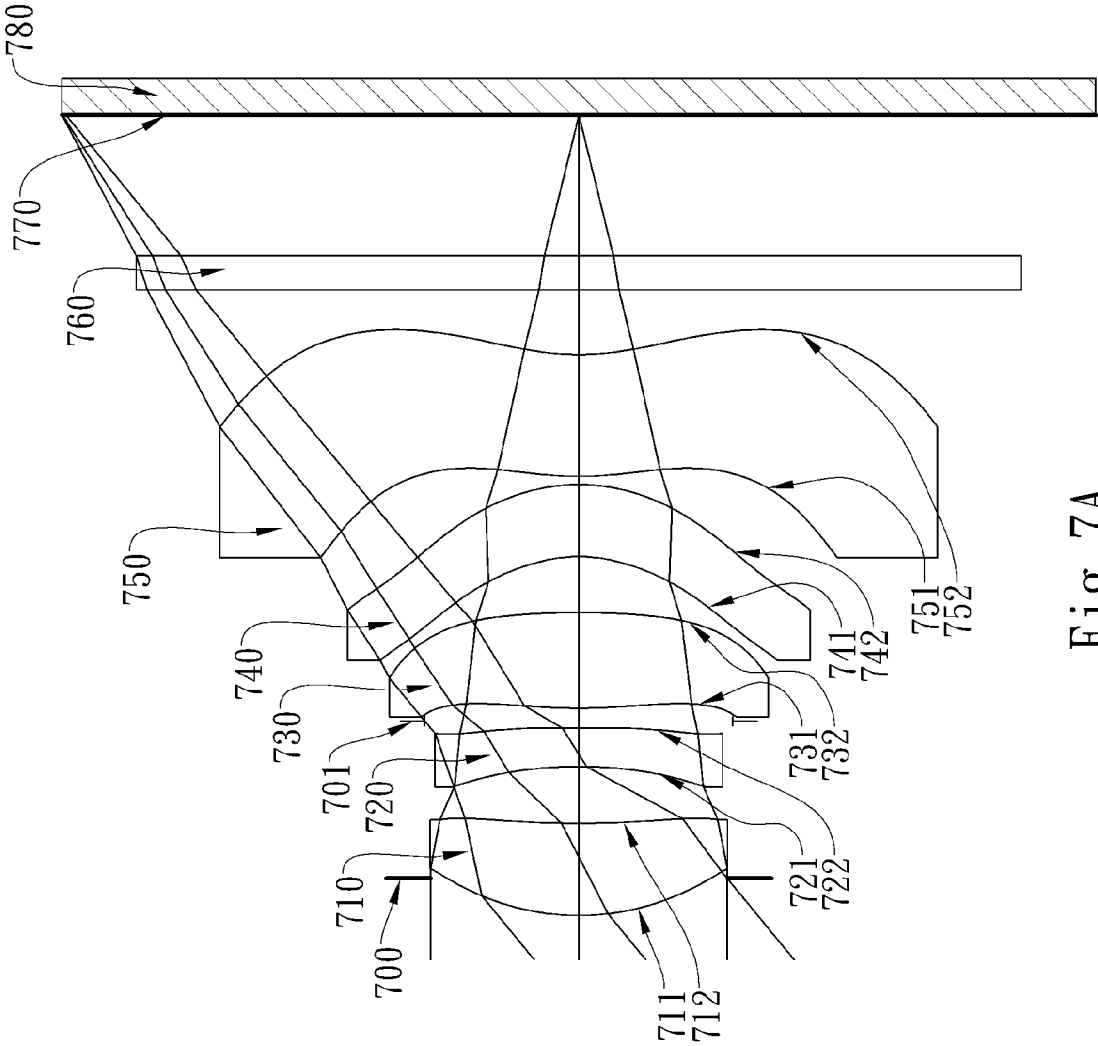


Fig. 7A

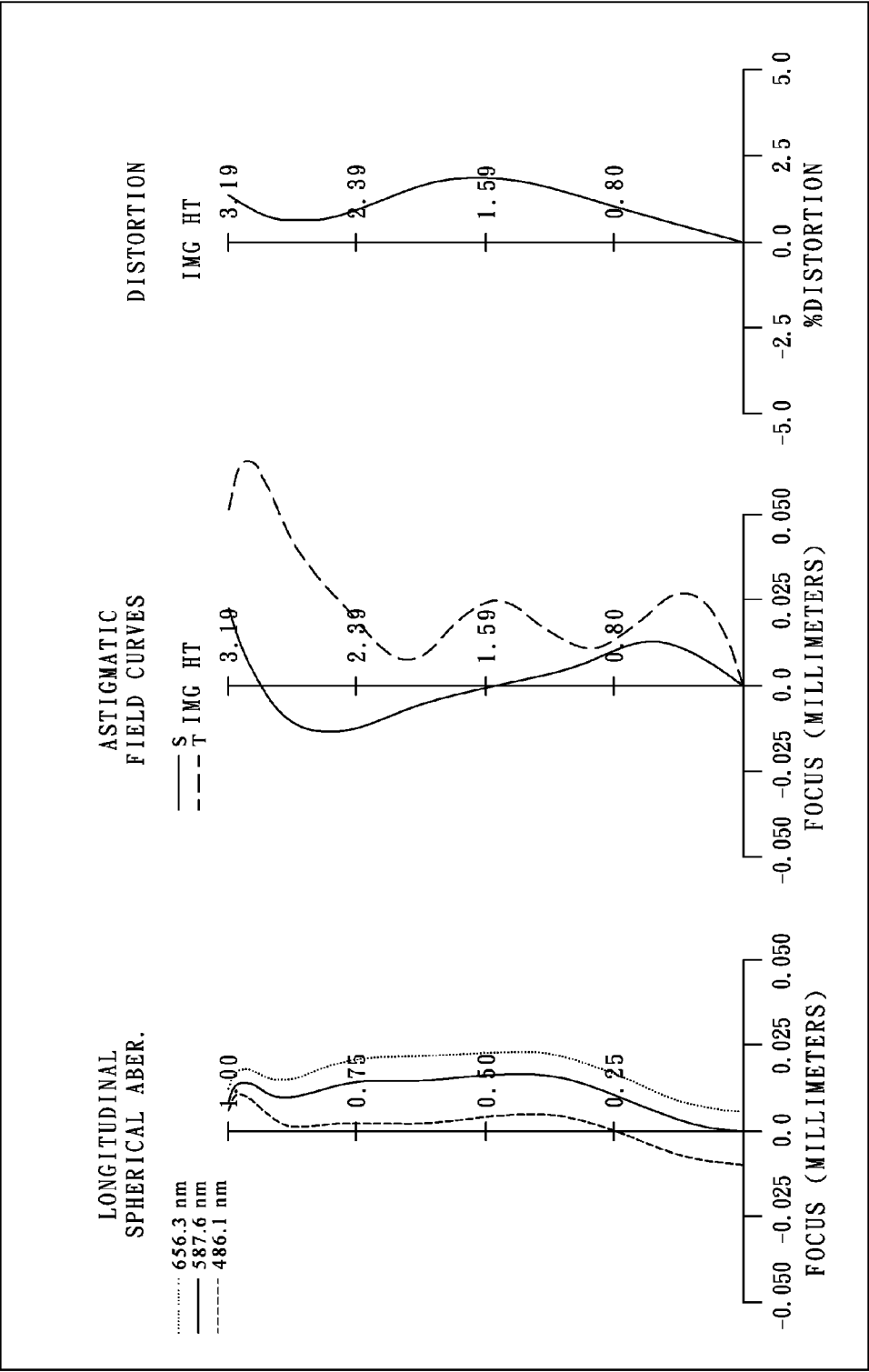


Fig. 7B

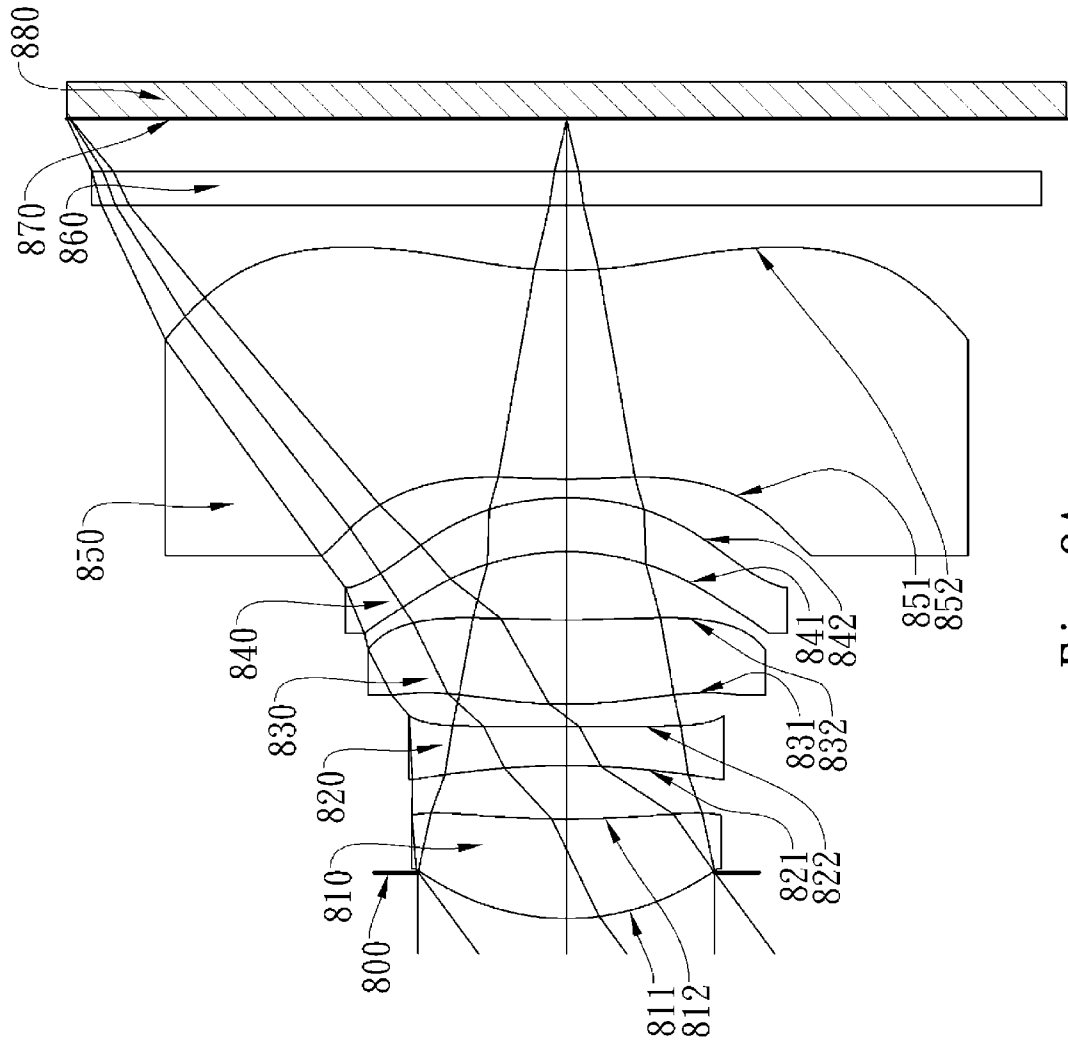


Fig. 8A

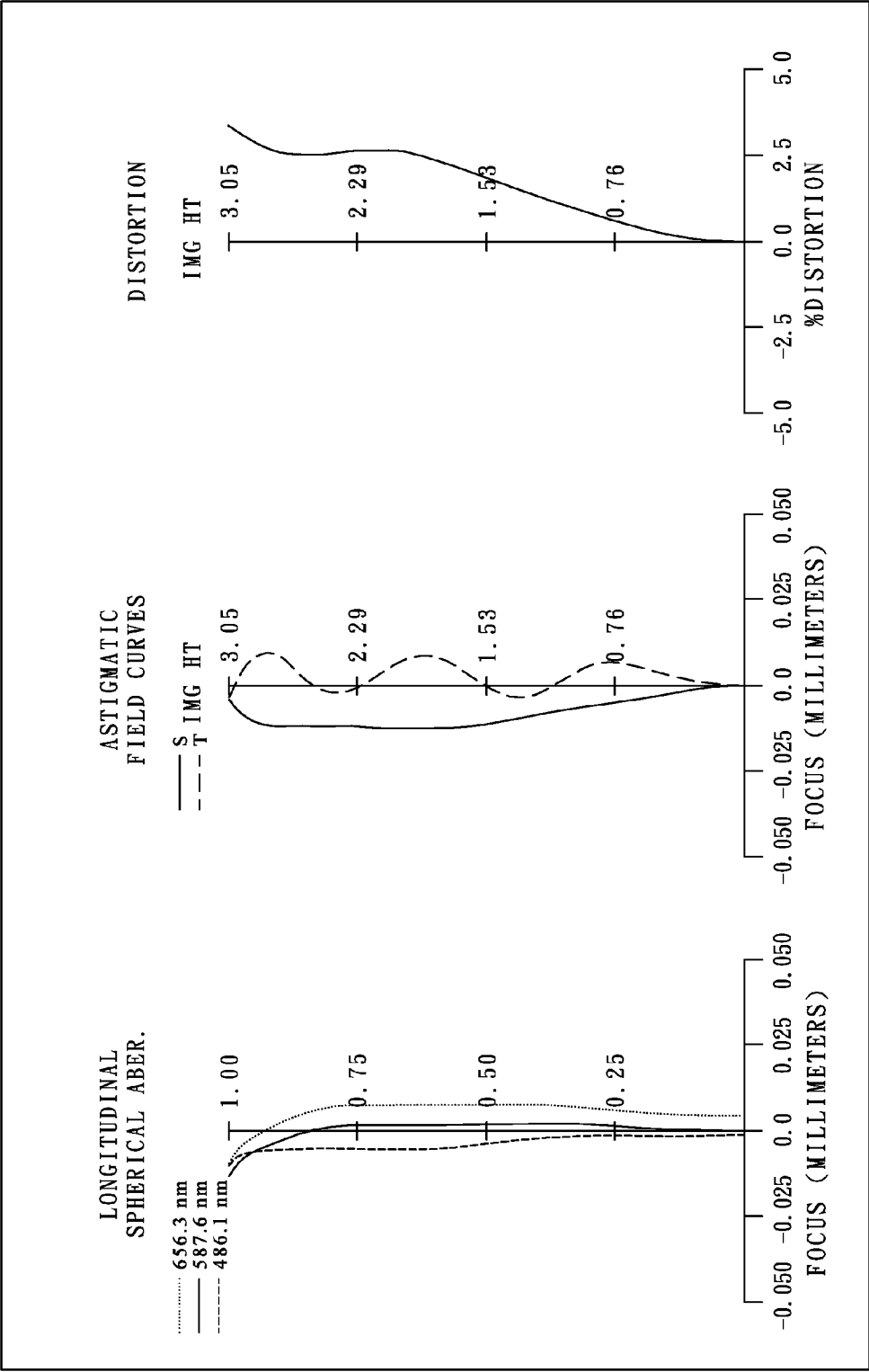


Fig. 8B

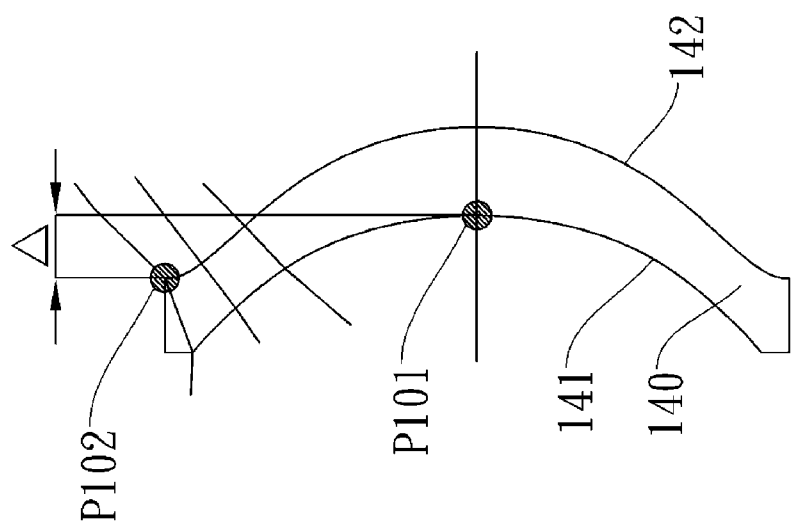


Fig. 9

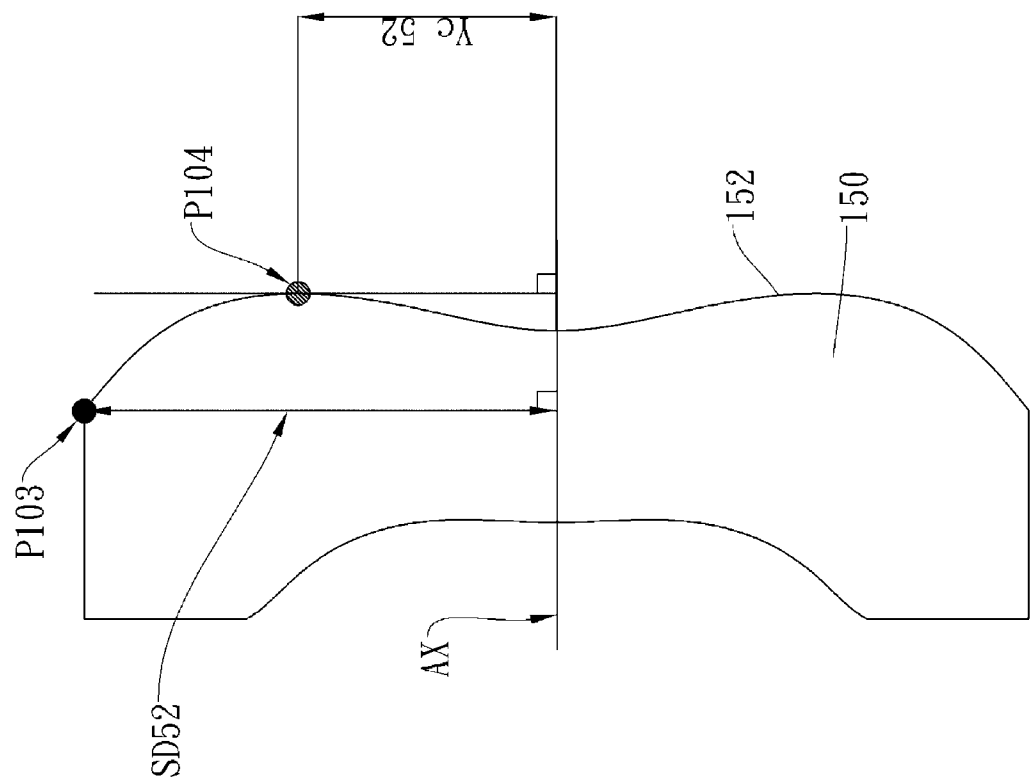


Fig. 10

1

IMAGE CAPTURING OPTICAL LENS SYSTEM

RELATED APPLICATIONS

The application claims priority to Taiwan Application Serial Number 101149767, filed Dec. 25, 2012, which is herein incorporated by reference.

BACKGROUND

1. Technical Field

The present invention relates to an image capturing optical lens system. More particularly, the present invention relates to a compact image capturing optical lens system applicable to portable electronic products.

2. Description of Related Art

In recent years, with the popularity of mobile products with camera functionalities, the demand of optical lens systems is increasing. The sensor of a conventional photographing camera is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide-Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical lens systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical lens systems featuring better image quality.

A conventional and compact optical lens system employed in a portable electronic product such as the one disclosed in the U.S. Pat. No. 8,179,470, mainly adopts a four-element lens structure. Due to the popularity of mobile products with high-end specifications, such as smart phones and PDAs (Personal Digital Assistants), the requirements for smaller pixel size and better image quality have been increasing. However, the conventional four-element lens structure cannot satisfy these requirements of the compact optical lens system.

Although other conventional optical lens systems with five-element lens structure such as the one disclosed in the U.S. Pat. No. 8,000,030 enhance image quality and resolving power, it still reside with unsolved problems. Since the arrangement of the lens elements which are close to the image side of the optical lens elements is not proper regarding the refractive index, it is not favorable for obtaining more of the image coverage. Furthermore, it is also not favorable for keeping the optical lens system compact due to the size of barrel. Therefore, a compact optical lens system which can be applied to the portable electronic products with larger refractive index on the image side of the optical lens system and with more of image coverage under limited peripheral regions of lens elements is called for solving the problems stated above.

SUMMARY

According to one aspect of the present disclosure, an image capturing optical lens system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The first lens element with positive refractive power has a convex object-side surface. The second lens element with negative refractive power has a concave object-side surface. The third lens element has refractive power. The fourth lens element with refractive power has a concave object-side surface and a convex image-side surface. The fifth lens element with refractive power has a concave image-side surface, wherein an object-side surface and the image-side

2

surface of the fifth lens element are aspheric, and the fifth lens element has at least one inflection point on the image-side surface thereof. When a refractive index of the fifth lens element is $N5$, the following relationship is satisfied:

$$1.60 < N5 < 1.75.$$

According to another aspect of the present disclosure, an image capturing optical lens system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The first lens element with positive refractive power has a convex object-side surface. The second lens element has negative refractive power. The third lens element has refractive power. The fourth lens element with refractive power has a concave object-side surface and a convex image-side surface. The fifth lens element with refractive power has a concave image-side surface, wherein an object-side surface and the image-side surface of the fifth lens element are aspheric, and the fifth lens element has at least one inflection point on the image-side surface thereof. When a refractive index of the fifth lens element is $N5$, a focal length of the image capturing optical lens system is f , and a focal length of the fourth lens element is $f4$, the following relationships are satisfied:

$$1.60 < N5 < 1.75; \text{ and}$$

$$-1.0 < f/f4 < 0.40.$$

According to still another aspect of the present disclosure, an image capturing optical lens system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The first lens element with positive refractive power has a convex object-side surface. The second lens element with negative refractive power has a concave object-side surface. The third lens element with positive refractive power has a convex object-side surface. The fourth lens element with refractive power has a convex image-side surface. The fifth lens element with refractive power has a concave image-side surface, wherein an object-side surface and the image-side surface of the fifth lens element are aspheric, and the fifth lens element has at least one inflection point on the image-side surface thereof. When a refractive index of the fifth lens element is $N5$, a central thickness of the fifth lens element is $CT5$, and a curvature radius of an object-side surface of the second lens element is $R3$, the following relationships are satisfied:

$$1.60 < N5 < 1.75; \text{ and}$$

$$-0.60 < CT5/R3 < -0.05.$$

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

FIG. 1A is a schematic view of an image capturing optical lens system according to the 1st embodiment of the present disclosure;

FIG. 1B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 1st embodiment;

FIG. 2A is a schematic view of an image capturing optical lens system according to the 2nd embodiment of the present disclosure;

3

FIG. 2B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 2nd embodiment;

FIG. 3A is a schematic view of an image capturing optical lens system according to the 3rd embodiment of the present disclosure;

FIG. 3B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 3rd embodiment;

FIG. 4A is a schematic view of an image capturing optical lens system according to the 4th embodiment of the present disclosure;

FIG. 4B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 4th embodiment;

FIG. 5A is a schematic view of an image capturing optical lens system according to the 5th embodiment of the present disclosure;

FIG. 5B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 5th embodiment;

FIG. 6A is a schematic view of an image capturing optical lens system according to the 6th embodiment of the present disclosure;

FIG. 6B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 6th embodiment;

FIG. 7A is a schematic view of an image capturing optical lens system according to the 7th embodiment of the present disclosure;

FIG. 7B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 7th embodiment;

FIG. 8A is a schematic view of an image capturing optical lens system according to the 8th embodiment of the present disclosure;

FIG. 8B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 8th embodiment;

FIG. 9 shows Δ of the image capturing optical lens system according to the fourth lens element of the 1st embodiment of the present disclosure; and

FIG. 10 shows SD52 and Yc52 of the image capturing optical lens system according to the fifth lens element of the 1st embodiment.

DETAILED DESCRIPTION

An image capturing optical lens system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element.

The first lens element has positive refractive power, so that the total track length of the image capturing optical lens system can be reduced. The first lens element has a convex object-side surface, so that it is favorable for further reducing the total track length.

The second lens element has negative refractive power, so that it is favorable for correcting the aberration generated by the first lens element. The second image second lens element has a concave object-side surface, so that it is favorable for further correcting the aberration.

The third lens element can have positive refractive power, so that it is favorable for reducing system sensitivity. The third lens element can have a convex object-side surface, so that it is favorable for further reducing the system sensitivity so as to increase the manufacturing yield rate.

4

The fourth lens element can have negative refractive power, so that it is favorable for correcting the aberration and reducing the system sensitivity. The fourth lens element can have a concave object-side surface and a convex image-side surface, so that the astigmatism of the image capturing optical lens system can be further corrected.

The fifth lens element with refractive power can have a convex object-side surface and has a concave image-side surface, so that it is favorable for correcting the astigmatism. The fifth lens element has at least one inflection point on the image-side surface, so that it is favorable for reducing the angle at which the incident light projects onto an image sensor from the off-axis so as to increase the responding efficiency of the image sensor and to further correct the aberration of the off-axis.

When a refractive index of the fifth lens element is N_5 , the following relationship is satisfied: $1.60 < N_5 < 1.75$. Therefore, the image side of the image capturing optical lens system can have a larger refractive index which is favorable for obtaining more of the image coverage under limited peripheral regions of the lens elements.

When a focal length of the image capturing optical lens system is f , and a focal length of the fourth lens element is f_4 , the following relationship is satisfied: $-1.0 < f/f_4 < -0.70$. It is favorable for reducing the system sensitivity and correcting the aberration so as to improve image quality. Preferably, the following relationship is further satisfied: $-1.0 < f/f_4 < -0.40$.

When a central thickness of the fifth lens element is CT_5 , and a curvature radius of the object-side surface of the second lens element is R_3 , the following relationship is satisfied: $-0.60 < CT_5/R_3 < -0.05$. Therefore, it is favorable for correcting the aberration and increasing the moldability and homogeneity for plastic lens elements during the injection molding process so as to achieve superior image quality.

When the refractive index of the fifth lens element is N_5 , and a refractive index of the first lens element is N_1 , the following relationship is satisfied: $0.06 < N_5 - N_1 < 0.25$. Therefore, it is favorable for the lens elements which are close to the image side of the image capturing optical lens system obtaining larger refractive index so as to acquire more of the image coverage under limited peripheral regions of the lens elements.

When a distance in parallel with an optical axis from an axial vertex on the object-side surface of the fourth lens element to a maximum effective diameter position on the image-side surface of the fourth lens element is Δ (When the distance towards the image side of the image capturing optical lens system is positive; when the distance towards the object side of the image capturing optical lens system is negative.), and a central thickness of the fourth lens element is CT_4 , the following relationship is satisfied: $-2.0 < \Delta/CT_4 < -0.20$. Therefore, the fourth lens element will not be excessively curved which is favorable for manufacturing and shaping the lens elements so as to reduce the required space for assembling the lens elements to keep the image capturing optical lens system compact.

When the central thickness of the fourth lens element is CT_4 , and the central thickness of the fifth lens element is CT_5 , the following relationship is satisfied: $0.20 < CT_4/CT_5 < 0.65$. Therefore, it is favorable for arranging the thickness of the lens elements and for providing favorable moldability and homogeneity for plastic lens elements during the injection molding process.

When a vertical distance between a non-axial critical point on the image-side surface of the fifth lens element and the optical axis is $Yc52$, wherein the non-axial critical point refers to this critical point is not on the optical axis, and the focal

5

length of the image capturing optical lens system is f , the following relationship is satisfied: $0.1 < Yc52/f < 0.7$. It is favorable for correcting the aberration of the off-axis region.

When a vertical distance between a maximum effective diameter position on the image-side surface of the fifth lens element and the optical axis is $SD52$, and a maximum image height of the image capturing optical lens system is $ImgH$ (half of a diagonal length of an effective photosensitive area of an image sensor), the following relationship is satisfied: $0.60 < SD52/ImgH < 0.77$. Therefore, it is favorable for manufacturing and shaping the lens elements and keeping the image capturing optical lens system compact.

When at least three lens elements with refractive power have refractive index larger than 1.60, it is favorable for balancing the arrangement of the refractive index of the image capturing optical lens system and obtaining more of the image coverage under limited peripheral regions of lens elements.

When a curvature radius of the image-side surface of the fifth lens element is $R10$, and a curvature radius of the object-side surface of the fifth lens element is $R9$, the following relationship is satisfied: $-0.20 < R10/R9 < 0.90$. Therefore, it is favorable for correcting the astigmatism so as to improve the image quality and resolving power.

When an Abbe number of the fourth lens element is $V4$, an Abbe number of the fifth lens element is $V5$, and an Abbe number of the first lens element is $V1$, the following relationship is satisfied: $0.7 < (V4+V5)/V1 < 1.0$. Therefore, it is favorable for correcting the chromatic aberration of the image capturing optical lens system.

According to the image capturing optical lens system of the present disclosure, the lens elements thereof can be made of plastic or glass material. When the lens elements are made of glass material, the distribution of the refractive power of the image capturing optical lens system may be more flexible to design. When the lens elements are made of plastic material, the manufacturing costs can be effectively reduced. Furthermore, the surfaces of each lens element can be aspheric, so that it is easier to make the surface into non-spherical shapes. As a result, more controllable variables are obtained, and the aberration is reduced, as well as the number of required lens elements can be reduced while constructing an optical lens system. Therefore, the total track length of the image capturing optical lens system can also be reduced.

According to the image capturing optical lens system of the present disclosure, there can be at least one stop provided, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is for eliminating the stray light and thereby improving the image resolution thereof. In the present image capturing optical lens system, an aperture stop can be configured as a front stop or a middle stop. A front stop can provide a longer distance between an exit pupil of the system and an image plane and which improves the image-sensing efficiency of an image sensor. A middle stop is favorable for enlarging the field of view of the system and thereby provides a wider field of view for the same.

According to the image capturing optical lens system of the present disclosure, each of an object-side surface and an image-side surface of every lens element has a paraxial region and a peripheral region. The paraxial region refers to the region of the surface where light rays travel close to an optical axis and the peripheral region refers to the region of the surface where light rays travel away from the optical axis. Particularly, when a lens element has a convex surface, it indicates that the surface is convex at the paraxial region and when the lens element has a concave surface, it indicates that the surface is concave at the paraxial region.

6

FIG. 9 shows a distance in parallel with the optical axis from an axial vertex **P101** on the object-side surface **141** of the fourth lens element **140** to a maximum effective diameter position **P102** on the image-side surface **142** of the fourth lens element **140** is Δ . When the distance towards the image side of the image capturing optical lens system is positive; when the distance towards the object side of the image capturing optical lens system is negative.

Critical point is a non-axial point of the lens surface where its tangent is perpendicular to the optical axis. FIG. 10 shows a vertical distance between a non-axial critical point **P104** on the image-side surface **152** of the fifth lens element **150** and the optical axis AX is $Yc52$. Most important of all, the critical point **P104** is not on the optical axis AX . In addition, a vertical distance between a maximum effective diameter position **P103** on the image-side surface **152** of the fifth lens element **150** and the optical axis AX is $SD52$.

According to the image capturing optical lens system of the present disclosure, the image capturing optical lens system is featured with good correction ability and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices and tablets.

According to the above description of the present disclosure, the following 1st-8th specific embodiments are provided for further explanation.

1st Embodiment

FIG. 1A is a schematic view of an image capturing optical lens system according to the 1st embodiment of the present disclosure. FIG. 1B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 1st embodiment. In FIG. 1A, the image capturing optical lens system includes five lens elements with refractive power, in order from an object side to an image side, an aperture stop **100**, a first lens element **110**, a second lens element **120**, a third lens element **130**, a fourth lens element **140**, a fifth lens element **150**, an IR-cut filter **160**, an image plane **170**, and an image sensor **180**.

The first lens element **110** with positive refractive power has a convex object-side surface **111** and a convex image-side surface **112**. The first lens element **110** is made of plastic material and has the object-side surface **111** and the image-side surface **112** being aspheric.

The second lens element **120** with negative refractive power has a concave object-side surface **121** and a concave image-side surface **122**. The second lens element **120** is made of plastic material and has the object-side surface **121** and the image-side surface **122** being aspheric.

The third lens element **130** with positive refractive power has a convex object-side surface **131** and a convex image-side surface **132**. The third lens element **130** is made of plastic material and has the object-side surface **131** and the image-side surface **132** being aspheric.

The fourth lens element **140** with positive refractive power has a concave object-side surface **141** and a convex image-side surface **142**. The fourth lens element **140** is made of plastic material and has the object-side surface **141** and the image-side surface **142** being aspheric.

The fifth lens element **150** with negative refractive power has a convex object-side surface **151** and a concave image-side surface **152**. The fifth lens element **150** is made of plastic material and has the object-side surface **151** and the image-

side surface **152** being aspheric. Furthermore, the fifth lens element **150** has at least one inflection point on the image-side surface **152** thereof.

The refractive index of the second lens element **120**, the fourth lens element **140** and the fifth lens element **150** are all larger than 1.6. Furthermore, the IR-cut filter **160** made of glass material is located between the fifth lens element **150** and the image plane **170**, and will not affect a focal length of the image capturing optical lens system.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (Ai) \times (Y^i),$$

wherein,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the distance from the point on the curve of the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

Ai is the i-th aspheric coefficient.

In the image capturing optical lens system according to the 1st embodiment, when a focal length of the image capturing optical lens system is f, an f-number of the image capturing optical lens system is Fno, and half of the maximal field of view of the image capturing optical lens system is HFOV, these parameters have the following values:

$$f=3.73\text{mm};$$

$$Fno=2.08; \text{ and}$$

$$HFOV=39.1\text{degrees}.$$

In the image capturing optical lens system according to the 1st embodiment, when a refractive index of the fifth lens element **150** is N5, the following relationship is satisfied: N5=1.640.

In the image capturing optical lens system according to the 1st embodiment, when the refractive index of the fifth lens element **150** is N5, and a refractive index of the first lens element **110** is N1, the following relationship is satisfied: N5-N1=0.096.

In the image capturing optical lens system according to the 1st embodiment, when an Abbe number of the fourth lens

element **140** is V4, an Abbe number of the fifth lens element **150** is V5, and an Abbe number of the first lens element **110** is V1, the following relationship is satisfied: (V4+V5)/V1=0.87.

In the image capturing optical lens system according to the 1st embodiment, when a central thickness of the fourth lens element **140** is CT4, and a central thickness of the fifth lens element **150** is CT5, the following relationship is satisfied: CT4/CT5=0.37.

In the image capturing optical lens system according to the 1st embodiment, when the central thickness of the fifth lens element **150** is CT5, and a curvature radius of the object-side surface **121** of the second lens element **120** is R3, the following relationship is satisfied: CT5/R3=-0.14.

In the image capturing optical lens system according to the 1st embodiment, when a curvature radius of the image-side surface **152** of the fifth lens element **150** is R10, and a curvature radius of the object-side surface **151** of the fifth lens element **150** is R9, the following relationship is satisfied: R10/R9=0.40.

In the image capturing optical lens system according to the 1st embodiment, when the focal length of the image capturing optical lens system is f, and a focal length of the fourth lens element **140** is f4, the following relationship is satisfied: f/f4=0.38.

In the image capturing optical lens system according to the 1st embodiment, when a distance in parallel with the optical axis from an axial vertex P101 (FIG. 9) on the object-side surface **141** of the fourth lens element **140** to a maximum effective diameter position P102 (FIG. 9) on the image-side surface **142** of the fourth lens element **140** is Δ, and the central thickness of the fourth lens element **140** is CT4, the following relationship is satisfied: Δ/CT4=-0.71.

In the image capturing optical lens system according to the 1st embodiment, when a vertical distance between a non-axial critical point P104 (FIG. 10) on the image-side surface **152** of the fifth lens element **150** and the optical axis AX is Yc52, wherein the critical point P104 is not on the optical axis AX (FIG. 10), and the focal length of the image capturing optical lens system is f, the following relationship is satisfied: Yc52/f=0.36.

In the image capturing optical lens system according to the 1st embodiment, when a vertical distance between a maximum effective diameter position P103 (FIG. 10) on the image-side surface **152** of the fifth lens element **150** and the optical axis AX (FIG. 10) is SD52, and a maximum image height of the image capturing optical lens system is ImgH, the following relationship is satisfied: SD52/ImgH=0.766.

The detailed optical data of the 1st embodiment are shown in Table 1, and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment								
f = 3.73 mm, Fno = 2.08, HFOV = 39.1 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.210					
2	Lens 1	1.844 (ASP)	0.635	Plastic	1.544	55.9	3.18	
3		-24.259 (ASP)	0.161					
4	Lens 2	-6.882 (ASP)	0.240	Plastic	1.650	21.4	-4.75	
5		5.667 (ASP)	0.203					
6	Lens 3	3.807 (ASP)	0.547	Plastic	1.535	55.7	6.13	
7		-22.564 (ASP)	0.447					
8	Lens 4	-1.578 (ASP)	0.371	Plastic	1.614	25.6	9.86	
9		-1.364 (ASP)	0.154					

TABLE 1-continued

1st Embodiment								
$f = 3.73$ mm, $Fno = 2.08$, $HFOV = 39.1$ deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
10	Lens 5	3.834 (ASP)	0.990	Plastic	1.640	23.3	-4.72	
11		1.519 (ASP)	0.400					
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.448					
14	Image	Plano	—					

Note:

Reference wavelength (d-line) is 587.6 nm.

TABLE 2

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-6.4121E+00	-9.0000E+01	-9.1845E+00	1.5146E+01	-5.8887E+01
A4 =	1.2093E-01	-1.3185E-02	-6.5661E-02	-1.7547E-01	-5.8649E-02
A6 =	-8.4032E-02	2.1920E-02	2.2160E-01	3.6264E-01	-1.0726E-01
A8 =	1.0925E-01	-7.7627E-02	-4.3882E-01	-6.1553E-01	1.5177E-01
A10 =	-1.8041E-01	6.6639E-02	4.8761E-01	6.2209E-01	-1.7670E-01
A12 =	1.7662E-01	-1.3382E-01	-4.5115E-01	-4.0012E-01	7.7825E-02
A14 =	-8.4454E-02	7.0479E-02	2.0069E-01	1.2886E-01	
Surface #	7	8	9	10	11
k =	2.0000E+01	-1.7605E+01	-8.8958E+00	-1.3912E+00	-7.9626E+00
A4 =	-5.2885E-02	-3.9298E-01	-5.1236E-01	-3.8394E-01	-7.2477E-02
A6 =	-9.1217E-02	7.3642E-01	8.1986E-01	4.1315E-01	2.6529E-02
A8 =	6.6376E-02	-1.5232E+00	-1.1583E+00	-4.6517E-01	-8.5693E-03
A10 =	-2.9937E-02	1.6491E+00	9.6605E-01	3.8632E-01	1.9673E-03
A12 =	8.1273E-04	-8.4285E-01	-3.9797E-01	-1.9843E-01	-3.3717E-04
A14 =		1.6342E-01	6.5946E-02	5.3782E-02	3.4763E-05
A16 =				-5.7670E-03	-1.4739E-06

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-14 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A1-A16 represent the aspheric coefficients ranging from the 1st order to the 16th order. This information related to Table 1 and Table 2 applies also to the Tables for the remaining embodiments, and so an explanation in this regard will not be provided again.

2nd Embodiment

FIG. 2A is a schematic view of an image capturing optical lens system according to the 2nd embodiment of the present disclosure. FIG. 2B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 2nd embodiment. In FIG. 2A, the image capturing optical lens system includes five lens elements with refractive power, in order from an object side to an image side, an aperture stop 200, a first lens element 210, a second lens element 220, a third lens element 230, a fourth lens element 240, a fifth lens element 250, an IR-cut filter 260, an image plane 270, and an image sensor 280.

The first lens element 210 with positive refractive power has a convex object-side surface 211 and a convex image-side surface 212. The first lens element 210 is made of plastic material and has the object-side surface 211 and the image-side surface 212 being aspheric.

The second lens element 220 with negative refractive power has a concave object-side surface 221 and a concave image-side surface 222. The second lens element 220 is made of plastic material and has the object-side surface 221 and the image-side surface 222 being aspheric.

The third lens element 230 with positive refractive power has a convex object-side surface 231 and a convex image-side surface 232. The third lens element 230 is made of plastic material and has the object-side surface 231 and the image-side surface 232 being aspheric.

The fourth lens element 240 with positive refractive power has a concave object-side surface 241 and a convex image-side surface 242. The fourth lens element 240 is made of plastic material and has the object-side surface 241 and the image-side surface 242 being aspheric.

The fifth lens element 250 with negative refractive power has a convex object-side surface 251 and a concave image-side surface 252. The fifth lens element 250 is made of plastic material and has the object-side surface 251 and the image-side surface 252 being aspheric. Furthermore, the fifth lens element 250 has at least one inflection point on the image-side surface 252 thereof.

The IR-cut filter 260 made of glass material is located between the fifth lens element 250 and the image plane 270, and will not affect a focal length of the image capturing optical lens system.

The detailed optical data of the 2nd embodiment are shown in Table 3, and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment							
f = 3.75 mm, Fno = 2.08, HFOV = 39.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.238				
2	Lens 1	1.748 (ASP)	0.604	Plastic	1.544	55.9	3.20
3		-317.089 (ASP)	0.138				
4	Lens 2	-15.320 (ASP)	0.240	Plastic	1.650	21.4	-5.53
5		4.717 (ASP)	0.262				
6	Lens 3	5.074 (ASP)	0.603	Plastic	1.535	55.7	9.18
7		-147.096 (ASP)	0.364				
8	Lens 4	-2.227 (ASP)	0.472	Plastic	1.583	30.2	5.08
9		-1.372 (ASP)	0.199				
10	Lens 5	5.464 (ASP)	0.870	Plastic	1.640	23.3	-3.37
11		1.450 (ASP)	0.400				
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
13		Plano	0.442				
14	Image	Plano	—				

Note:
Reference wavelength (d-line) is 587.6 nm.

TABLE 4

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-5.8736E+00	2.0000E+01	2.0000E+01	1.4936E+01	-9.0000E+01
A4 =	1.3597E-01	-1.0285E-02	-7.5820E-02	-1.4931E-01	-7.0856E-02
A6 =	-8.0289E-02	4.6200E-02	2.5217E-01	3.2479E-01	-8.6337E-02
A8 =	1.1225E-01	-8.5551E-02	-4.6358E-01	-5.6948E-01	1.3212E-01
A10 =	-1.7680E-01	5.4664E-02	5.0292E-01	6.2597E-01	-1.7947E-01
A12 =	1.7640E-01	-1.0162E-01	-4.1504E-01	-4.1229E-01	9.9593E-02
A14 =	-8.3496E-02	4.9100E-02	1.6660E-01	1.3959E-01	
Surface #	7	8	9	10	11
k =	-1.0000E+01	-4.2943E+01	-8.9291E+00	-1.0290E+00	-8.0846E+00
A4 =	-6.9795E-02	-3.9339E-01	-5.0980E-01	-3.8000E-01	-8.1561E-02
A6 =	-1.0254E-01	7.4355E-01	8.2643E-01	4.1834E-01	3.1509E-02
A8 =	7.3168E-02	-1.5370E+00	-1.1566E+00	-4.6371E-01	-9.3893E-03
A10 =	-3.6940E-02	1.6503E+00	9.6455E-01	3.8558E-01	1.9077E-03
A12 =	6.8491E-03	-8.3623E-01	-3.9963E-01	-1.9896E-01	-3.2232E-04
A14 =		1.6116E-01	6.5588E-02	5.3687E-02	3.6431E-05
A16 =				-5.6854E-03	-1.7307E-06

In the image capturing optical lens system according to the 2nd embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment. Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following relationships:

f [mm]	3.75	CT5/R3	-0.06
Fno	2.08	R10/R9	0.27
HFOV [deg.]	39.0	f/f4	0.74
N5	1.640	Δ/CT4	-0.39
N5-N1	0.096	Yc52/f	0.35
(V4 + V5)/V1	0.96	SD52/ImgH	0.759
CT4/CT5	0.54		

3rd Embodiment

FIG. 3A is a schematic view of an image capturing optical lens system according to the 3rd embodiment of the present disclosure. FIG. 3B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image captur-

ing optical lens system according to the 3rd embodiment. In FIG. 3A, the image capturing optical lens system includes five lens elements with refractive power, in order from an object side to an image side, an aperture stop **300**, a first lens element **310**, a second lens element **320**, a third lens element **330**, a fourth lens element **340**, a fifth lens element **350**, an IR-cut filter **360**, an image plane **370**, and an image sensor **380**.

The first lens element **310** with positive refractive power has a convex object-side surface **311** and a concave image-side surface **312**. The first lens element **310** is made of plastic material and has the object-side surface **311** and the image-side surface **312** being aspheric.

The second lens element **320** with negative refractive power has a concave object-side surface **321** and a concave image-side surface **322**. The second lens element **320** is made of plastic material and has the object-side surface **321** and the image-side surface **322** being aspheric.

The third lens element **330** with positive refractive power has a convex object-side surface **331** and a concave image-side surface **332**. The third lens element **330** is made of plastic material and has the object-side surface **331** and the image-side surface **332** being aspheric.

13

The fourth lens element **340** with positive refractive power has a concave object-side surface **341** and a convex image-side surface **342**. The fourth lens element **340** is made of plastic material and has the object-side surface **341** and the image-side surface **342** being aspheric.

The fifth lens element **350** with negative refractive power has a concave object-side surface **351** and a concave image-side surface **352**. The fifth lens element **350** is made of plastic material and has the object-side surface **351** and the image-side surface **352** being aspheric. Furthermore, the fifth lens element **350** has at least one inflection point on the image-side surface **352** thereof.

The IR-cut filter **360** made of glass material is located between the fifth lens element **350** and the image plane **370**, and will not affect a focal length of the image capturing optical lens system.

The detailed optical data of the 3rd embodiment are shown in Table 5, and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment								
f = 3.75 mm, Fno = 2.08, HFOV = 38.9 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.286					
2	Lens 1	1.568 (ASP)	0.547	Plastic	1.544	55.9	3.21	
3		13.300 (ASP)	0.152					
4	Lens 2	-8.052 (ASP)	0.252	Plastic	1.650	21.4	-6.28	
5		8.372 (ASP)	0.284					
6	Lens 3	5.366 (ASP)	0.542	Plastic	1.535	55.7	11.04	
7		56.550 (ASP)	0.326					
8	Lens 4	-2.607 (ASP)	0.500	Plastic	1.543	56.5	4.35	
9		-1.323 (ASP)	0.386					
10	Lens 5	-81.633 (ASP)	0.870	Plastic	1.640	23.3	-2.61	
11		1.714 (ASP)	0.400					
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.249					
14	Image	Plano	—					

Note:

Reference wavelength (d-line) is 587.6 nm.

TABLE 6

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-4.4498E+00	-8.0053E+01	1.4186E+01	-1.3249E+01	-9.0000E+01
A4 =	1.5189E-01	-2.9293E-02	-1.0477E-01	-1.2803E-01	-1.2325E-01
A6 =	-6.9419E-02	5.8190E-02	3.1204E-01	3.8247E-01	-6.9384E-02
A8 =	1.1837E-01	-5.6380E-02	-4.3039E-01	-5.0639E-01	1.2464E-01
A10 =	-1.7109E-01	8.6861E-02	5.1047E-01	6.0145E-01	-1.9263E-01
A12 =	1.8012E-01	-9.9797E-02	-4.3661E-01	-4.4551E-01	1.2063E-01
A14 =	-8.2931E-02	-2.5824E-03	1.2770E-01	1.7592E-01	
Surface #	7	8	9	10	11
k =	-1.0000E+01	-5.2611E+01	-8.3778E+00	-9.0000E+01	-1.0628E+01
A4 =	-9.4927E-02	-4.1769E-01	-5.3403E-01	-3.5089E-01	-8.1494E-02
A6 =	-1.1203E-01	7.3992E-01	8.3408E-01	4.1225E-01	3.1958E-02
A8 =	6.9812E-02	-1.5322E+00	-1.1581E+00	-4.6313E-01	-9.3526E-03
A10 =	-3.2905E-02	1.6505E+00	9.6414E-01	3.8546E-01	1.8701E-03
A12 =	7.1911E-03	-8.3470E-01	-3.9921E-01	-1.9913E-01	-3.2335E-04
A14 =		1.6250E-01	6.5909E-02	5.3654E-02	3.7277E-05
A16 =				-5.6539E-03	-1.7457E-06

In the image capturing optical lens system according to the 3rd embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st

14

embodiment with corresponding values for the 3rd embodiment. Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following relationships:

f [mm]	3.75	CT5/R3	-0.11
Fno	2.08	R10/R9	-0.02
HFOV [deg.]	38.9	f/f4	0.86
N5	1.640	Δ/CT4	-0.39
N5-N1	0.096	Yc52/f	0.33
(V4 + V5)/V1	1.43	SD52/ImgH	0.768
CT4/CT5	0.57		

4th Embodiment

FIG. 4A is a schematic view of an image capturing optical lens system according to the 4th embodiment of the present disclosure. FIG. 4B shows spherical aberration curves, astig-

65 matic field curves and a distortion curve of the image capturing optical lens system according to the 4th embodiment. In FIG. 4A, the image capturing optical lens system includes

15

five lens elements with refractive power, in order from an object side to an image side, a first lens element **410**, an aperture stop **400**, a second lens element **420**, a third lens element **430**, a fourth lens element **440**, a fifth lens element **450**, an IR-cut filter **460**, an image plane **470**, and an image sensor **480**.

The first lens element **410** with positive refractive power has a convex object-side surface **411** and a concave image-side surface **412**. The first lens element **410** is made of plastic material and has the object-side surface **411** and the image-side surface **412** being aspheric.

The second lens element **420** with negative refractive power has a concave object-side surface **421** and a concave image-side surface **422**. The second lens element **420** is made of plastic material and has the object-side surface **421** and the image-side surface **422** being aspheric.

The third lens element **430** with positive refractive power has a convex object-side surface **431** and a convex image-side

16

The fourth lens element **440** with positive refractive power has a concave object-side surface **441** and a convex image-side surface **442**. The fourth lens element **440** is made of plastic material and has the object-side surface **441** and the image-side surface **442** being aspheric.

The fifth lens element **450** with negative refractive power has a convex object-side surface **451** and a concave image-side surface **452**. The fifth lens element **450** is made of plastic material and has the object-side surface **451** and the image-side surface **452** being aspheric. Furthermore, the fifth lens element **450** has at least one inflection point on the image-side surface **452** thereof.

The IR-cut filter **460** made of glass material is located between the fifth lens element **450** and the image plane **470**, and will not affect a focal length of the image capturing optical lens system.

The detailed optical data of the 4th embodiment are shown in Table 7, and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment								
f = 3.74 mm, Fno = 2.20, HFOV = 39.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Lens 1	1.614 (ASP)	0.605	Plastic	1.535	55.7	3.47	
2		10.857 (ASP)	0.044					
3	Ape. Stop	Plano	0.127					
4	Lens 2	-14.275 (ASP)	0.240	Plastic	1.650	21.4	-6.67	
5		6.265 (ASP)	0.269					
6	Lens 3	5.478 (ASP)	0.572	Plastic	1.530	55.8	10.20	
7		-406.669 (ASP)	0.305					
8	Lens 4	-2.055 (ASP)	0.465	Plastic	1.543	56.5	14.91	
9		-1.770 (ASP)	0.227					
10	Lens 5	2.280 (ASP)	0.870	Plastic	1.640	23.3	-9.12	
11		1.395 (ASP)	0.400					
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.501					
14	Image	Plano	—					

Note:

Reference wavelength (d-line) is 587.6 nm.

TABLE 8

Aspheric Coefficients					
Surface #	1	2	4	5	6
k =	-4.9677E+00	-4.8104E+01	2.0000E+01	-2.0346E+01	-8.9540E+01
A4 =	1.5026E-01	-3.0616E-02	-1.2228E-01	-1.3222E-01	-1.0876E-01
A6 =	-7.2422E-02	5.6310E-02	3.1166E-01	3.7375E-01	-6.0349E-02
A8 =	1.1043E-01	-6.0851E-02	-4.3364E-01	-5.0461E-01	1.1683E-01
A10 =	-1.7505E-01	7.9875E-02	5.1222E-01	6.0448E-01	-2.0124E-01
A12 =	1.8364E-01	-1.0503E-01	-4.2812E-01	-4.4781E-01	1.2263E-01
A14 =	-7.7077E-02	3.8120E-02	1.3929E-01	1.6328E-01	
Surface #	7	8	9	10	11
k =	-5.4297E+00	-3.2859E+01	-5.8203E+00	-1.9950E+01	-7.0161E+00
A4 =	-7.9152E-02	-4.2058E-01	-5.3244E-01	-3.6492E-01	-8.8152E-02
A6 =	-1.1415E-01	7.4065E-01	8.3483E-01	4.1054E-01	3.2690E-02
A8 =	6.6995E-02	-1.5285E+00	-1.1591E+00	-4.6263E-01	-9.3802E-03
A10 =	-3.1424E-02	1.6528E+00	9.6348E-01	3.8568E-01	1.8571E-03
A12 =	8.1995E-03	-8.3412E-01	-3.9932E-01	-1.9910E-01	-3.2388E-04
A14 =		1.6218E-01	6.5998E-02	5.3640E-02	3.7491E-05
A16 =				-5.6618E-03	-1.6860E-06

surface **432**. The third lens element **430** is made of plastic material and has the object-side surface **431** and the image-side surface **432** being aspheric.

In the image capturing optical lens system according to the 4th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st

17

embodiment with corresponding values for the 4th embodiment. Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following relationships:

f [mm]	3.74	CT5/R3	-0.06
Fno	2.20	R10/R9	0.61
HFOV [deg.]	39.0	f/f4	0.25
N5	1.640	Δ/CT4	-0.52
N5-N1	0.105	Yc52/f	0.35
(V4 + V5)/V1	1.43	SD52/ImgH	0.746
CT4/CT5	0.53		

5th Embodiment

FIG. 5A is a schematic view of an image capturing optical lens system according to the 5th embodiment of the present disclosure. FIG. 5B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 5th embodiment. In FIG. 5A, the image capturing optical lens system includes five lens elements with refractive power, in order from an object side to an image side, an aperture stop 500, a first lens element 510, a second lens element 520, a stop 501, a third lens element 530, a fourth lens element 540, a fifth lens element 550, an IR-cut filter 560, an image plane 570, and an image sensor 580.

The first lens element 510 with positive refractive power has a convex object-side surface 511 and a concave image-side surface 512. The first lens element 510 is made of plastic material and has the object-side surface 511 and the image-side surface 512 being aspheric.

18

The second lens element 520 with negative refractive power has a concave object-side surface 521 and a concave image-side surface 522. The second lens element 520 is made of plastic material and has the object-side surface 521 and the image-side surface 522 being aspheric.

The third lens element 530 with positive refractive power has a convex object-side surface 531 and a convex image-side surface 532. The third lens element 530 is made of plastic material and has the object-side surface 531 and the image-side surface 532 being aspheric.

The fourth lens element 540 with negative refractive power has a concave object-side surface 541 and a convex image-side surface 542. The fourth lens element 540 is made of plastic material and has the object-side surface 541 and the image-side surface 542 being aspheric.

The fifth lens element 550 with positive refractive power has a convex object-side surface 551 and a concave image-side surface 552. The fifth lens element 550 is made of plastic material and has the object-side surface 551 and the image-side surface 552 being aspheric. Furthermore, the fifth lens element 550 has at least one inflection point on the image-side surface 552 thereof.

The refractive index of the second lens element 520, the fourth lens element 540 and the fifth lens element 550 are all larger than 1.6. Furthermore, the IR-cut filter 560 made of glass material is located between the fifth lens element 550 and the image plane 570, and will not affect a focal length of the image capturing optical lens system.

The detailed optical data of the 5th embodiment are shown in Table 9, and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment								
f = 3.83 mm, Fno = 2.08, HFOV = 39.3 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.213					
2	Lens 1	1.578 (ASP)	0.539	Plastic	1.544	55.9	4.00	
3		5.054 (ASP)	0.376					
4	Lens 2	-4.208 (ASP)	0.240	Plastic	1.634	23.8	-3.65	
5		5.243 (ASP)	0.070					
6	stop	Plano	0.000					
7	Lens 3	1.883 (ASP)	0.547	Plastic	1.544	55.9	2.77	
8		-6.741 (ASP)	0.419					
9	Lens 4	-0.765 (ASP)	0.367	Plastic	1.634	23.8	-10.53	
10		-1.025 (ASP)	0.050					
11	Lens 5	2.227 (ASP)	0.914	Plastic	1.634	23.8	1563.63	
12		1.877 (ASP)	0.400					
13	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
14		Plano	0.808					
15	Image	Plano	—					

Note:

Reference wavelength (d-line) is 587.6 nm.

Half of the diameter of the stop at Surface 6 is 0.915 mm.

TABLE 10

Aspheric Coefficients					
Surface #	2	3	4	5	7
k =	-4.0732E+00	-1.0000E+00	5.4853E+00	-9.0000E+01	-1.6518E+01
A4 =	1.2751E-01	-3.6131E-02	-1.5012E-01	-3.8374E-01	-8.8850E-02
A6 =	-6.7596E-02	-1.1795E-02	2.5329E-01	6.0974E-01	-9.3131E-02
A8 =	9.9086E-02	-6.9140E-02	-3.0569E-01	-5.6745E-01	1.5160E-01
A10 =	-1.9511E-01	8.1485E-02	4.0578E-01	5.5582E-01	-1.0983E-01

TABLE 10-continued

Aspheric Coefficients					
A12 =	2.0939E-01	-1.4282E-01	-4.3966E-01	-4.0931E-01	-7.1125E-03
A14 =	-1.1437E-01	6.7290E-02	2.1806E-01	1.5674E-01	
Surface #	8	9	10	11	12
k =	6.8236E+00	-4.9218E+00	-6.5844E+00	-1.5695E+00	-1.1226E+01
A4 =	1.2007E-01	-2.8030E-01	-4.8641E-01	-4.2548E-01	-8.0298E-02
A6 =	-2.5117E-01	6.1678E-01	8.3317E-01	4.4299E-01	2.6739E-02
A8 =	4.9408E-02	-1.4062E+00	-1.1613E+00	-4.8831E-01	-8.9039E-03
A10 =	6.0164E-02	1.6653E+00	9.7074E-01	3.8931E-01	1.9378E-03
A12 =	-3.8329E-02	-8.8929E-01	-3.9408E-01	-1.9694E-01	-3.0162E-04
A14 =		1.7564E-01	6.0891E-02	5.3693E-02	2.9401E-05
A16 =				-5.8392E-03	-1.1682E-06

In the image capturing optical lens system according to the 5th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment. Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following relationships:

f [mm]	3.83	CT5/R3	-0.22
Fno	2.08	R10/R9	0.84
HFOV [deg.]	39.3	f/f4	-0.36
N5	1.634	Δ/CT4	-0.77
N5-N1	0.090	Yc52/f	0.29
(V4 + V5)/V1	0.85	SD52/ImgH	0.691
CT4/CT5	0.40		

6th Embodiment

FIG. 6A is a schematic view of an image capturing optical lens system according to the 6th embodiment of the present disclosure. FIG. 6B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 6th embodiment. In FIG. 6A, the image capturing optical lens system includes five lens elements with refractive power, in order from an object side to an image side, an aperture stop **600**, a first lens element **610**, a stop **601**, a second lens element **620**, a third lens element **630**, a fourth lens element **640**, a fifth lens element **650**, an IR-cut filter **660**, an image plane **670**, and an image sensor **680**.

The first lens element **610** with positive refractive power has a convex object-side surface **611** and a concave image-side surface **612**. The first lens element **610** is made of plastic

material and has the object-side surface **611** and the image-side surface **612** being aspheric.

The second lens element **620** with negative refractive power has a concave object-side surface **621** and a convex image-side surface **622**. The second lens element **620** is made of plastic material and has the object-side surface **621** and the image-side surface **622** being aspheric.

The third lens element **630** with positive refractive power has a convex object-side surface **631** and a convex image-side surface **632**. The third lens element **630** is made of plastic material and has the object-side surface **631** and the image-side surface **632** being aspheric.

The fourth lens element **640** with positive refractive power has a concave object-side surface **641** and a convex image-side surface **642**. The fourth lens element **640** is made of plastic material and has the object-side surface **641** and the image-side surface **642** being aspheric.

The fifth lens element **650** with negative refractive power has a convex object-side surface **651** and a concave image-side surface **652**. The fifth lens element **650** is made of plastic material and has the object-side surface **651** and the image-side surface **652** being aspheric. Furthermore, the fifth lens element **650** has at least one inflection point on the image-side surface **652** thereof.

The refractive index of the second lens element **620**, the fourth lens element **640** and the fifth lens element **650** are all larger than 1.6. Furthermore, the IR-cut filter **660** made of glass material is located between the fifth lens element **650** and the image plane **670**, and will not affect a focal length of the image capturing optical lens system.

The detailed optical data of the 6th embodiment are shown in Table 11, and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment f = 3.81 mm, Fno = 2.08, HFOV = 39.4 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.265					
2	Lens 1	1.609 (ASP)	0.649	Plastic	1.544	55.9	3.84	
3		6.006 (ASP)	0.069					
4	stop	Plano	0.256					
5	Lens 2	-2.681 (ASP)	0.241	Plastic	1.634	23.8	-4.35	
6		-100.000 (ASP)	0.112					
7	Lens 3	2.859 (ASP)	0.596	Plastic	1.544	55.9	3.34	
8		-4.611 (ASP)	0.406					
9	Lens 4	-0.930 (ASP)	0.474	Plastic	1.634	23.8	5.81	

TABLE 11-continued

6th Embodiment							
f = 3.81 mm, Fno = 2.08, HFOV = 39.4 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
10		-0.889 (ASP)	0.050				
11	Lens 5	2.644 (ASP)	0.590	Plastic	1.634	23.8	-3.97
12		1.178 (ASP)	0.500				
13	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
14		Plano	0.790				
15	Image	Plano	—				

Note:

Reference wavelength (d-line) is 587.6 nm.

Half of the diameter of the stop at Surface 4 is 0.900 mm.

TABLE 12

Aspheric Coefficients					
Surface #	2	3	5	6	7
k =	-4.4008E+00	-1.0000E+00	3.9550E-01	-9.0000E+01	-3.1417E+01
A4 =	1.3505E-01	-2.2794E-02	-1.1233E-01	-2.8196E-01	-9.3500E-02
A6 =	-5.0508E-02	-5.2846E-02	1.2368E-01	5.2154E-01	-1.5880E-03
A8 =	7.9729E-02	5.0548E-02	-1.1073E-01	-6.1721E-01	8.3109E-02
A10 =	-1.9439E-01	-7.2576E-02	1.6393E-01	6.5195E-01	-1.2871E-01
A12 =	2.6066E-01	-1.1791E-01	-4.2934E-01	-4.9295E-01	2.9824E-02
A14 =	-1.4071E-01	8.6809E-02	2.8949E-01	1.8654E-01	
Surface #	8	9	10	11	12
k =	4.0009E+00	-1.8397E+00	-3.1506E+00	-3.7738E+00	-8.1126E+00
A4 =	1.1872E-02	1.1838E-01	-2.0098E-01	-2.7829E-01	-1.0907E-01
A6 =	-1.8912E-01	-5.7638E-01	1.5263E-01	1.7152E-01	5.0957E-02
A8 =	1.8351E-01	6.7030E-01	-1.8971E-01	-8.9855E-02	-2.2083E-02
A10 =	-9.9426E-02	-2.9498E-01	1.9579E-01	2.4902E-02	5.9007E-03
A12 =	2.1156E-02	5.3565E-02	-8.3819E-02	-3.7111E-03	-9.5221E-04
A14 =		-4.8606E-03	1.2224E-02	3.3923E-04	8.2497E-05
A16 =				-1.4769E-05	-2.8401E-06

In the image capturing optical lens system according to the 6th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment. Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following relationships:

f [mm]	3.81	CT5/R3	-0.22
Fno	2.08	R10/R9	0.45
HFOV [deg.]	39.4	f/f4	0.66
N5	1.634	Δ /CT4	-0.76
N5-N1	0.090	Yc52/f	0.30
(V4 + V5)/V1	0.85	SD52/ImgH	0.690
CT4/CT5	0.80		

7th Embodiment

FIG. 7A is a schematic view of an image capturing optical lens system according to the 7th embodiment of the present disclosure. FIG. 7B shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 7th embodiment. In FIG. 7A, the image capturing optical lens system includes five lens elements with refractive power, in order from an object side to an image side, an aperture stop 700, a first lens element 710, a second lens element 720, a stop 701, a third

lens element 730, a fourth lens element 740, a fifth lens element 750, an IR-cut filter 760, an image plane 770, and an image sensor 780.

The first lens element 710 with positive refractive power has a convex object-side surface 711 and a concave image-side surface 712. The first lens element 710 is made of plastic material and has the object-side surface 711 and the image-side surface 712 being aspheric.

The second lens element 720 with negative refractive power has a concave object-side surface 721 and a convex image-side surface 722. The second lens element 720 is made of plastic material and has the object-side surface 721 and the image-side surface 722 being aspheric.

The third lens element 730 with positive refractive power has a convex object-side surface 731 and a convex image-side surface 732. The third lens element 730 is made of plastic material and has the object-side surface 731 and the image-side surface 732 being aspheric.

The fourth lens element 740 with negative refractive power has a concave object-side surface 741 and a convex image-side surface 742. The fourth lens element 740 is made of plastic material and has the object-side surface 741 and the image-side surface 742 being aspheric.

The fifth lens element 750 with negative refractive power has a convex object-side surface 751 and a concave image-side surface 752. The fifth lens element 750 is made of plastic material and has the object-side surface 751 and the image-side surface 752 being aspheric. Furthermore, the fifth lens element 750 has at least one inflection point on the image-side surface 752 thereof.

23

The refractive index of the second lens element **720**, the fourth lens element **740** and the fifth lens element **750** are all larger than 1.6. Furthermore, the IR-cut filter **760** made of glass material is located between the fifth lens element **750** and the image plane **770**, and will not affect a focal length of the image capturing optical lens system.

The detailed optical data of the 7th embodiment are shown in Table 13, and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment							
f = 3.81 mm, Fno = 2.08, HFOV = 39.4 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.228				
2	Lens 1	1.544 (ASP)	0.568	Plastic	1.544	55.9	3.76
3		5.463 (ASP)	0.350				
4	Lens 2	-3.063 (ASP)	0.240	Plastic	1.634	23.8	-4.99
5		-100.000 (ASP)	0.041				
6	stop	Plano	0.080				
7	Lens 3	3.036 (ASP)	0.590	Plastic	1.544	55.9	3.61
8		-5.182 (ASP)	0.346				
9	Lens 4	-0.878 (ASP)	0.445	Plastic	1.634	23.8	-18.93
10		-1.134 (ASP)	0.050				
11	Lens 5	1.953 (ASP)	0.751	Plastic	1.634	23.8	-28.92
12		1.501 (ASP)	0.400				
13	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
14		Plano	0.870				
15	Image	Plano	—				

Note:

Reference wavelength (d-line) is 587.6 nm.

Half of the diameter of the stop at Surface 6 is 0.950 mm.

TABLE 14

Aspheric Coefficients					
Surface #	2	3	4	5	7
k =	-3.9282E+00	-1.0000E+00	2.6830E+00	2.0000E+01	-3.9770E+01
A4 =	1.3527E-01	-3.4006E-02	-1.4691E-01	-3.3391E-01	-1.3237E-01
A6 =	-5.7250E-02	-3.4010E-02	2.5137E-01	5.9594E-01	-3.2488E-02
A8 =	8.6064E-02	-1.9376E-02	-2.4743E-01	-5.5588E-01	1.3786E-01
A10 =	-1.9452E-01	3.1780E-02	3.6086E-01	5.5907E-01	-1.5922E-01
A12 =	2.4712E-01	-1.5873E-01	-5.0342E-01	-4.4146E-01	8.6774E-03
A14 =	-1.4672E-01	9.3581E-02	2.8811E-01	1.9053E-01	
Surface #	8	9	10	11	12
k =	1.6132E+01	-4.3925E+00	-6.4166E+00	-4.6622E+00	-5.8871E+00
A4 =	2.1204E-02	-1.2459E-01	-4.1211E-01	-3.5262E-01	-1.3814E-01
A6 =	-2.1503E-01	-1.8947E-01	4.7048E-01	2.3340E-01	7.1013E-02
A8 =	1.5835E-01	1.7594E-01	-4.9709E-01	-1.2218E-01	-3.0015E-02
A10 =	-5.7390E-02	9.4370E-02	3.7810E-01	3.5698E-02	8.1020E-03
A12 =	3.1625E-03	-1.0733E-01	-1.4296E-01	-5.7084E-03	-1.3370E-03
A14 =		2.0486E-02	1.9974E-02	5.6094E-04	1.1990E-04
A16 =				-2.6300E-05	-4.3115E-06

55

In the image capturing optical lens system according to the 7th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment. Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following relationships:

f [mm]	3.81	CT5/R3	-0.25
Fno	2.08	R10/R9	0.77

65

24

-continued

HFOV [deg.]	39.4	f/f4	-0.20
N5	1.634	Δ /CT4	-0.74
N5-N1	0.090	Yc52/f	0.30
(V4 + V5)/V1	0.85	SD52/ImgH	0.696
CT4/CT5	0.59		

8th Embodiment

FIG. **8A** is a schematic view of an image capturing optical lens system according to the 8th embodiment of the present disclosure. FIG. **8B** shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing optical lens system according to the 8th embodiment. In FIG. **8A**, the image capturing optical lens system includes five lens elements with refractive power, in order from an object side to an image side, an aperture stop **800**, a first lens element **810**, a second lens element **820**, a third lens element

25

830, a fourth lens element **840**, a fifth lens element **850**, an IR-cut filter **860**, an image plane **870**, and an image sensor **880**.

The first lens element **810** with positive refractive power has a convex object-side surface **811** and a concave image-side surface **812**. The first lens element **810** is made of plastic material and has the object-side surface **811** and the image-side surface **812** being aspheric.

The second lens element **820** with negative refractive power has a concave object-side surface **821** and a concave image-side surface **822**. The second lens element **820** is made of plastic material and has the object-side surface **821** and the image-side surface **822** being aspheric.

The third lens element **830** with positive refractive power has a convex object-side surface **831** and a concave image-side surface **832**. The third lens element **830** is made of plastic material and has the object-side surface **831** and the image-side surface **832** being aspheric.

The fourth lens element **840** with negative refractive power has a concave object-side surface **841** and a convex image-

26

The fifth lens element **850** with negative refractive power has a convex object-side surface **851** and a concave image-side surface **852**. The fifth lens element **850** is made of plastic material and has the object-side surface **851** and the image-side surface **852** being aspheric. Furthermore, the fifth lens element **850** has at least one inflection point on the image-side surface **852** thereof.

The refractive index of the second lens element **820**, the fourth lens element **840** and the fifth lens element **850** are all larger than 1.6. Furthermore, the IR-cut filter **860** made of glass material is located between the fifth lens element **850** and the image plane **870**, and will not affect a focal length of the image capturing optical lens system.

The detailed optical data of the 8th embodiment are shown in Table 15, and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment								
f = 4.00 mm, Fno = 2.20, HFOV = 36.4 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.284					
2	Lens 1	1.516 (ASP)	0.611	Plastic	1.535	56.3	3.65	
3		5.845 (ASP)	0.327					
4	Lens 2	-5.170 (ASP)	0.241	Plastic	1.650	21.4	-6.09	
5		17.144 (ASP)	0.136					
6	Lens 3	2.334 (ASP)	0.519	Plastic	1.544	55.9	5.55	
7		9.503 (ASP)	0.420					
8	Lens 4	-1.415 (ASP)	0.330	Plastic	1.634	23.8	-21.47	
9		-1.722 (ASP)	0.115					
10	Lens 5	4.017 (ASP)	1.280	Plastic	1.632	23.4	-11.38	
11		2.259 (ASP)	0.400					
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.324					
14	Image	Plano	—					

Note:

Reference wavelength (d-line) is 587.6 nm.

TABLE 16

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-3.9818E+00	-1.7061E+00	-3.5824E+01	2.0000E+01	-2.1017E+01
A4 =	1.4246E-01	-1.5545E-02	-7.6653E-02	-1.8522E-01	-5.3491E-02
A6 =	-7.1906E-02	-1.4913E-03	2.1414E-01	4.7986E-01	-6.6256E-02
A8 =	1.0599E-01	-7.3633E-02	-3.8298E-01	-5.9342E-01	1.5585E-01
A10 =	-1.7844E-01	8.8421E-02	4.9672E-01	6.1006E-01	-1.7957E-01
A12 =	1.7677E-01	-1.3078E-01	-4.5093E-01	-4.0835E-01	7.1914E-02
A14 =	-8.5634E-02	5.3982E-02	1.8606E-01	1.6102E-01	
Surface #	7	8	9	10	11
k =	-7.0867E+00	-1.3570E+01	-1.6213E+01	-1.0910E+00	-8.1233E+00
A4 =	-1.7531E-02	-3.3757E-01	-5.1320E-01	-3.9343E-01	-6.7665E-02
A6 =	-1.2248E-01	7.1389E-01	8.3645E-01	4.1669E-01	2.5260E-02
A8 =	7.0431E-02	-1.4907E+00	-1.1552E+00	-4.6568E-01	-8.5036E-03
A10 =	-4.2108E-03	1.6570E+00	9.6663E-01	3.8539E-01	2.0254E-03
A12 =	-1.1964E-02	-8.5475E-01	-3.9810E-01	-1.9879E-01	-3.4099E-04
A14 =		1.6452E-01	6.3441E-02	5.3846E-02	3.3748E-05
				-5.6563E-03	-1.3979E-06

side surface **842**. The fourth lens element **840** is made of plastic material and has the object-side surface **841** and the image-side surface **842** being aspheric.

In the image capturing optical lens system according to the 8th embodiment, the definitions of these parameters shown in the following table are the same as those stated in the 1st

embodiment with corresponding values for the 8th embodiment. Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following relationships:

f [mm]	4.00	CT5/R3	-0.25
Fno	2.20	R10/R9	0.56
HFOV [deg.]	36.4	f/f4	-0.19
N5	1.632	Δ /CT4	-0.67
N5-N1	0.097	Yc52/f	0.33
(V4 + V5)/V1	0.84	SD52/ImgH	0.807
CT4/CT5	0.26		

It is to be noted that TABLES 1-16 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. Therefore, any imaging lens system of the same structure is considered to be within the scope of the present disclosure even if it uses different data. The embodiments depicted above and the appended drawings are exemplary and are not intended to limit the scope of the present disclosure.

What is claimed is:

1. An image capturing optical lens system comprising five lens elements with refractive power, in order from an object side to an image side:

- a first lens element with positive refractive power having a convex object-side surface;
- a second lens element with negative refractive power having a concave object-side surface;
- a third lens element having refractive power;
- a fourth lens element with refractive power having a concave object-side surface and a convex image-side surface; and
- a fifth lens element with refractive power having a convex object-side surface and a concave image-side surface, wherein an object-side surface and the image-side surface of the fifth lens element are aspheric, and the fifth lens element has at least one inflection point on the image-side surface thereof;

wherein the image capturing optical lens system has a total of five lens elements with refractive power; and wherein a refractive index of the fifth lens element is N5, and the following relationship is satisfied:

$$1.60 < N5 < 1.75.$$

2. The image capturing optical lens system of claim 1, wherein the third lens element has positive refractive power.

3. The image capturing optical lens system of claim 2, wherein the third lens element has a convex object-side surface.

4. The image capturing optical lens system of claim 2, wherein the refractive index of the fifth lens element is N5, a refractive index of the first lens element is N1, and the following relationship is satisfied:

$$0.06 < N5 - N1 < 0.25.$$

5. The image capturing optical lens system of claim 2, wherein a focal length of the image capturing optical lens system is f, a focal length of the fourth lens element is f4, and the following relationship is satisfied:

$$-1.0 < f/f4 < 0.70.$$

6. The image capturing optical lens system of claim 2, wherein a distance in parallel with an optical axis from an axial vertex on the object-side surface of the fourth lens element to a maximum effective diameter position on the

image-side surface of the fourth lens element is Δ , a central thickness of the fourth lens element is CT4, and the following relationship is satisfied:

$$-2.0 < \Delta/CT4 < -0.20.$$

7. The image capturing optical lens system of claim 1, wherein a central thickness of the fifth lens element is CT5, a curvature radius of the object-side surface of the second lens element is R3, and the following relationship is satisfied:

$$-0.60 < CT5/R3 < -0.05.$$

8. The image capturing optical lens system of claim 7, wherein a central thickness of the fourth lens element is CT4, the central thickness of the fifth lens element is CT5, and the following relationship is satisfied:

$$0.20 < CT4/CT5 < 0.65.$$

9. The image capturing optical lens system of claim 7, wherein a focal length of the image capturing optical lens system is f, a focal length of the fourth lens element is f4, and the following relationship is satisfied:

$$-1.0 < f/f4 < 0.40.$$

10. The image capturing optical lens system of claim 7, wherein a vertical distance between a non-axial critical point on the image-side surface of the fifth lens element and an optical axis is Yc52, a focal length of the image capturing optical lens system is f, and the following relationship is satisfied:

$$0.1 < Yc52/f < 0.7.$$

11. The image capturing optical lens system of claim 1, wherein a vertical distance between a maximum effective diameter position on the image-side surface of the fifth lens element and an optical axis is SD52, a maximum image height of the image capturing optical lens system is ImgH, and the following relationship is satisfied:

$$0.60 < SD52/ImgH < 0.77.$$

12. An image capturing optical lens system comprising five lens elements with refractive power, in order from an object side to an image side:

- a first lens element with positive refractive power having a convex object-side surface;
- a second lens element with negative refractive power having a concave image-side surface;
- a third lens element having refractive power;
- a fourth lens element with refractive power having a concave object-side surface and a convex image-side surface; and
- a fifth lens element with refractive power having a concave image-side surface, wherein an object-side surface and the image-side surface of the fifth lens element are aspheric, and the fifth lens element has at least one inflection point on the image-side surface thereof;

wherein the image capturing optical lens system has a total of five lens elements with refractive power; and wherein a refractive index of the fifth lens element is N5, a focal length of the image capturing optical lens system is f, a focal length of the fourth lens element is f4, and the following relationships are satisfied:

$$1.60 < N5 < 1.75; \text{ and}$$

$$-1.0 < f/f4 < 0.40.$$

13. The image capturing optical lens system of claim 12, wherein the fourth lens element has negative refractive power.

29

14. The image capturing optical lens system of claim 12, wherein the refractive index of the fifth lens element is N5, a refractive index of the first lens element is N1, and the following relationship is satisfied:

$$0.06 < N5 - N1 < 0.25.$$

15. The image capturing optical lens system of claim 12, comprising at least three lens elements with refractive power having refractive index larger than 1.60.

16. The image capturing optical lens system of claim 12, wherein a curvature radius of the image-side surface of the fifth lens element is R10, a curvature radius of the object-side surface of the fifth lens element is R9, and the following relationship is satisfied:

$$-0.20 < R10/R9 < 0.90.$$

17. The image capturing optical lens system of claim 16, wherein a distance in parallel with an optical axis from an axial vertex on the object-side surface of the fourth lens element to a maximum effective diameter position on the image-side surface of the fourth lens element is Δ , a central thickness of the fourth lens element is CT4, and the following relationship is satisfied:

$$-2.0 < \Delta/CT4 < -0.20.$$

18. The image capturing optical lens system of claim 16, wherein a vertical distance between a maximum effective diameter position on the image-side surface of the fifth lens element and an optical axis is SD52, a maximum image height of the image capturing optical lens system is ImgH, and the following relationship is satisfied:

$$0.60 < SD52/ImgH < 0.77.$$

19. An image capturing optical lens system comprising five lens elements with refractive power, in order from an object side to an image side:

- a first lens element with positive refractive power having a convex object-side surface;
- a second lens element with negative refractive power having a concave object-side surface;
- a third lens element with positive refractive power having a convex object-side surface;
- a fourth lens element with refractive power having a convex image-side surface; and

30

a fifth lens element with refractive power having a concave image-side surface, wherein an object-side surface and the image-side surface of the fifth lens element are aspheric, and the fifth lens element has at least one inflection point on the image-side surface thereof;

wherein the image capturing optical lens system has a total of five lens elements with refractive power; and wherein a refractive index of the fifth lens element is N5, a central thickness of the fifth lens element is CT5, a curvature radius of an object-side surface of the second lens element is R3, and the following relationships are satisfied:

$$1.60 < N5 < 1.75; \text{ and}$$

$$-0.60 < CT5/R3 < -0.05.$$

20. The image capturing optical lens system of claim 19, wherein the object-side surface of the fifth lens element is convex, a central thickness of the fourth lens element is CT4, the central thickness of the fifth lens element is CT5, and the following relationship is satisfied:

$$0.20 < CT4/CT5 < 0.65.$$

21. The image capturing optical lens system of claim 19, comprising at least three lens elements with refractive power having refractive index larger than 1.60.

22. The image capturing optical lens system of claim 19, wherein a focal length of the image capturing optical lens system is f, a focal length of the fourth lens element is f4, and the following relationship is satisfied:

$$-1.0 < f/f4 < 0.70.$$

23. The image capturing optical lens system of claim 19, wherein an Abbe number of the fourth lens element is V4, an Abbe number of the fifth lens element is V5, an Abbe number of the first lens element is V1, and the following relationship is satisfied:

$$0.7 < (V4 + V5)/V1 < 1.0.$$

* * * * *